UNIVERSITY OF ASIA PACIFIC

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Course Title: Software Engineering Lab | Course Code: CSE 314 | Section: B2



SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

SerialSync

A Multi-Location Doctor Appointment Booking & Real-Time Serial/Queue Tracker with Home Sample Collection and Report Delivery [MERN Stack]

SUBMITTED BY:	SUBMITTED TO:
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Submission Date: 27 August 2026 | Academic Session: Fall 2026

Team Members & Project Roles

SerialSync was conceptualized, designed, specified, and implemented by a collaborative four-member software engineering team from the Department of Computer Science and Engineering at the University of Asia Pacific.



FM Maftahul Jannat (Student ID: 23201084)

Role: **Project Manager & Requirement Analyst**

Responsible for overall project scheduling, stakeholder survey administration (n=56), feasibility study, risk mitigation planning, sprint tracking, and quality assurance coordination across all project milestones.



Md Adil Hossain (Student ID: 23201102)

Role: **Team Lead & Full-Stack Architect**

Lead architectural designer for MERN stack integration, Socket.io real-time event pipeline, atomic queue locking algorithms, database schema design, and core SRS documentation synthesis.



Md Abu Rafe Mostak H. Fahim (Student ID: 23101059)

Role: **QA Lead & Database Designer**

Engineered Express.js REST API microservices, MongoDB Mongoose schema models, Redis caching pipelines, JWT authentication, and diagnostic report HMAC verification.



Rohan Hasan Khan (Student ID: 23201108)

Role: **Frontend Developer & Report Lead**

Developed responsive React 19 / Next.js user interfaces, TailwindCSS design system, Socket.io client state hooks, test case execution, and cross-browser responsiveness validation.

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GUI design revision (October 2026)

Figures 5.6 through 5.12 in Chapter 5 have been updated to match the team Figma file SerialSync Clinical Care Hub and the implemented Next.js 15 web application.

Figma source: figma.com/design/vR4DaYPMmYq2EMyNGTOyx1

Exported reference PNGs and screen-to-figure mapping: docs/SRS_GUI_FIGURES.md and docs/design/figma/ in the SerialSync repository.

Original wireframe captions in Chapter 5 remain valid for functional scope; only the visual mockups were replaced.

Executive Summary

SerialSync is an enterprise-grade, centralized web application engineered using the **MERN Stack** (MongoDB, Express.js, React 19, Node.js) along with Redis and Socket.io to resolve two chronic bottlenecks in modern healthcare delivery: **(1) multi-location specialist schedule fragmentation with opaque physical waiting room queues**, and **(2) manual diagnostic specimen logistics and delayed physical report delivery**. In South Asian developing healthcare ecosystems, medical specialists routinely practice across three to five independent hospitals, private consultation clinics, and diagnostic centers on distinct days of the week. Patients struggle with fragmented scheduling information, relying on uncoordinated phone calls or early-morning physical visits to secure a consultation serial. Upon arrival at the clinic, patients endure unpredictable two to four-hour waiting room delays because queue progression is strictly tracked on private paper registers inside the consultation chamber.

Furthermore, diagnostic laboratory workflows inflict severe physical overhead on patients. A patient requiring routine blood or pathology tests must commute physically to a diagnostic center for specimen handover and make a second physical trip days later simply to collect a paper report. For elderly patients, rural families traveling to urban medical hubs, and acute patients, this operational fragmentation creates severe physical, financial, and emotional burdens.

SerialSync addresses these structural challenges through a self-contained, multi-tenant digital ecosystem organized into three core functional modules:

1. **Multi-Location Schedule & Chamber Discovery Layer:** Empowers verified doctors to configure complex, weekly rotating itineraries across independent hospitals and diagnostic centers. Patients gain unified, filterable search access by clinical specialty, degrees, hospital affiliations, consultation fees, and available days, backed by atomic sequential serial allocation.
2. **Real-Time Live Queue Tracker & Assistant Control Console:** Integrates Socket.io WebSocket channels and Redis in-memory pub/sub to stream live consultation progression ('Now Serving: #12', estimated wait time, delays) directly to patient mobile screens. Simultaneously, physical queue assistants operate a responsive, one-tap control console that supports emergency walk-in serial insertion (e.g. #11B) without disrupting pre-booked patient sequences.
3. **Home Diagnostic Sample Collection & Digital Report Portal:** Allows patients to book home specimen collection with certified phlebotomists, track sample collection and lab analysis stages in real-time, and download tamper-proof, HMAC-SHA256 encrypted digital PDF test reports that can be shared with consulting physicians in one click.

Key Technological Highlights & Empirical Evaluation

To ensure uncompromising engineering quality, SerialSync is formulated strictly according to the Board of Accreditation for Engineering and Technical Education (BAETE) and Washington Accord Outcome-Based Education (OBE) guidelines. Supported by primary empirical feedback from **56 stakeholders** (41 patients, 4 practicing medical specialists, 3 queue assistants, and 8 administrators), this Software Requirements Specification (SRS) establishes the definitive technical roadmap, architectural blueprints, UML diagrams, complete REST/WebSocket API contracts, 50+ system test cases, and Complex Engineering Problem (CEP) justifications for SerialSync.

Evaluation Dimension	Empirical Baseline & Operational Metric	SerialSync System Delivery
Average Clinic Waiting Delay	120 – 240 minutes in physical lounge	< 15 minutes physical wait via remote Live Queue Tracker
Queue State Broadcast Latency	N/A (Opaque paper registers)	< 150 milliseconds via Socket.io WebSocket channels
Serial Concurrency Safety	Frequent double-booking via phone	100% race-free atomic Redis `INCRBY` token allocation
Emergency Walk-in Handling	Disrupts entire pre-booked list	Priority sub-indexing (#11B) maintaining monotonic serial order
Diagnostic Report Turnaround	2 – 3 days with 2 physical hospital trips	Same-day digital PDF delivery with SHA-256 integrity hash
Doctor Multi-Chamber Visibility	0% unified discovery across clinics	100% weekly rotating schedule transparency on single portal

Table: Operational Performance Benchmark & Healthcare Impact Matrix

1. Introduction

1.1 Goals and Objectives of the Project

In developing urban communities and metropolitan medical clusters, finding up-to-date specialist chamber itineraries, securing a reliable consultation serial, and tracking waiting room queue progression remain largely manual, informal, and inefficient. Patients routinely rely on fragmented telephone inquiries, early-morning physical line-ups, or uncoordinated clinic visits—channels that lack real-time visibility, automated quota enforcement, and structured delay notifications. **SerialSync** addresses these practical healthcare gaps by providing a unified, centralized web platform engineered with the **MERN Stack** (MongoDB, Express.js, React 19, Node.js) paired with Socket.io and Redis to coordinate multi-location specialist schedules, deliver sub-150ms live queue updates, and streamline home diagnostic specimen collection and digital report delivery.

To definitively modernize outpatient clinical coordination, SerialSync establishes six core objectives:

- **Multi-Location Specialist Itinerary Synchronization:** Empower verified medical practitioners to manage rotating chamber schedules across multiple hospitals, private clinics, and diagnostic centers with automated quota and conflict management.
- **Atomic Sequential Serial Allocation:** Provide race-free serial token assignment backed by Redis atomic counter increments (`INCRBY`) and MongoDB transactional sessions, preventing double-booking during peak morning booking rushes.
- **Real-Time Waiting Room Visibility via WebSockets:** Stream instantaneous consultation queue updates ('Now Serving: #12', estimated wait time, delay notices) to patient mobile browsers via Socket.io WebSocket channels backed by Redis Pub/Sub.
- **Chamber Operator Empowerment & Walk-in Integration:** Equip physical clinic queue assistants with an intuitive mobile control panel capable of one-tap serial advancement, delay broadcasting, and emergency walk-in slot injection (e.g. Serial #11B) that maintains numerical order without disrupting pre-booked sequences.
- **Home Diagnostic Specimen Logistics & Verification:** Connect patients with accredited diagnostic laboratories and certified phlebotomists for on-demand home sample collection and tamper-evident digital PDF report delivery via SHA-256 cryptographic hashes and time-expiring presigned HMAC URLs.
- **Self-Contained Multi-Tenant Architecture:** Deliver a standalone, zero-integration web platform that operates without requiring invasive, costly integrations with closed proprietary hospital Electronic Medical Record (EMR) systems.

1.2 Scope of the Work

A centralized web-based multi-location doctor appointment booking and real-time serial/queue tracking platform connects several classes of users: patients acting as appointment seekers or diagnostic clients, verified doctors configuring multi-facility chamber itineraries, physical queue assistants managing in-chamber patient progression, certified phlebotomists collecting biological specimens, and system administrators governing platform integrity. The system coordinates the complete outpatient journey—from specialist discovery, chamber selection, and serial booking to remote queue tracking, in-chamber consultation, home diagnostic sample collection, and digital report delivery.

In-Scope Functional Deliverables:

1. *User Account & Role-Based Access Control (RBAC):* Secure JWT authentication with HTTP-only cookies across Patients, Doctors, Assistants, Phlebotomists, and Administrators.
2. *Doctor Multi-Location Chamber Schedule Builder:* Weekly recurring itinerary builder with multi-hospital chamber time-slots, patient quotas, consultation fees, and automated conflict detection.
3. *Intelligent Doctor & Chamber Discovery:* Multi-criteria search engine allowing patients to filter by clinical specialty, degrees, hospital affiliation, geographic area, and consultation day.
4. *Atomic Serial Reservation:* High-concurrency slot booking engine with sequential serial number assignment and downloadable QR-coded digital appointment tickets.
5. *Live Queue Tracker (Patient View):* Real-time WebSocket-powered monitor rendering active serving serial, patients ahead, estimated wait time, delay notices, and audio-visual turn alerts.
6. *Queue Assistant Control Console:* Responsive, one-handed mobile control panel for 1-tap serial advancement, emergency walk-in slot injection (#11B), delay broadcasting, and 60-second undo buffers.
7. *Home Diagnostic Specimen Collection:* Test catalog selection, residential address validation, phlebotomist dispatch, and barcode specimen tracking.
8. *Digital Medical Report Portal:* Upload of digitally signed PDF lab reports to encrypted AWS S3 storage with SHA-256 integrity verification and 1-click sharing with consulting doctors.
9. *Administrative Governance:* Doctor BMDC registration verification, chamber activation approval, and platform operational analytics.

1.3 System Overview

SerialSync is organized into three major functional layers supported by a foundational security and moderation infrastructure: **(1) Multi-Location Scheduling & Serial Booking Module**, which manages doctor profiles, rotating hospital chamber timetables, slot quotas, search filters, and atomic token generation; **(2) Real-Time Queue Tracking & Chamber Console Module**, which orchestrates live queue state synchronization, priority walk-in interleaving, delay broadcasts, assistant console controls, and push notifications; **(3) Diagnostic Specimen Collection & Report Delivery Module**, which handles home collection booking, phlebotomist dispatch, specimen barcode tracking, S3 report storage, and HMAC verification; and **(4) Platform Moderation & Security Layer**, which provides BMDC registration verification, chamber approval workflows, RBAC authorization, and tamper-evident audit logs.

1.4 Structure of the Document

The entire document is divided into 9 chapters, described below:

• **Chapter 1:** Provides a thorough overview of the project, including goals, scope, system overview, and glossary of key technical terms.

• **Chapter 2:** Details the project management approach—individual contributions, process model selection, risk analysis, implementation constraints, hardware and software requirements, project timeline, estimated budget, and feasibility analysis.

• **Chapter 3:** Presents a comprehensive Software Requirements Specification (SRS) for both core modules, stakeholder analysis (n=56), and exhaustive UML models (Use Case, Activity, Swimlane, Class, Sequence, State, and Security).

• **Chapter 4:** Defines the multi-tier MERN system architecture and design rationale.

• **Chapter 5:** Details component-level design patterns, Data Flow Diagrams (DFD Levels 0, 1, 2), Entity Relationship Diagrams (ERD), and high-fidelity GUI wireframes.

• **Chapter 6:** Delivers the REST and WebSocket API documentation (6 modules) with automated pass and fail test cases.

• **Chapter 7:** Formulates the system testing suite (50+ test cases), Requirements Traceability Matrix (RTM), and sustainability plan.

• **Chapter 8:** Maps the project to the CSE 314 Complex Engineering Problem (CEP) knowledge, problem and activity attributes.

• **Chapter 9:** Concludes the document and lists reference links.

1.5 Glossary of Technical Terms

Term / Abbreviation	Definition and Context in SerialSync
BMDC	Bangladesh Medical and Dental Council. The statutory regulatory body that issues mandatory registration numbers to qualified physicians.
Chamber Schedule	A doctor's recurring weekly time slot at a specific clinic, hospital, or diagnostic center (e.g. Green Life Medical, Mon & Wed 6-9 PM).
Serial Number	A sequential integer token assigned to a patient upon appointment booking that defines consultation order.
Walk-in Serial	An on-site emergency or manual booking inserted into an active queue (e.g. Serial #11B) without disrupting subsequent pre-booked patients.
MERN Stack	MongoDB (NoSQL Document DB), Express.js (REST API backend), React 19 (SPA frontend), Node.js (JavaScript runtime engine).
Socket.io	A bi-directional, event-driven WebSocket abstraction library enabling sub-100ms real-time queue status streaming to patient browsers.
HMAC-SHA256	Hash-based Message Authentication Code used to digitally sign diagnostic report download links, preventing unauthorized URL tampering.
Phlebotomist	A certified clinical laboratory agent dispatched to patient residences to collect biological blood and pathology samples.
RBAC	Role-Based Access Control. Security mechanism restricting system access to authorized roles (Patient, Doctor, Assistant, Admin).
Redis Pub/Sub	Publish/Subscribe messaging paradigm used in Redis to broadcast queue state changes across multiple Node.js cluster worker processes.
Presigned URL	A time-limited cryptographic URL granting temporary read access to private objects stored in AWS S3 buckets without exposing credentials.

Chapter 2: Project Management Plan and Feasibility Analysis

2.1 Project Organization

2.1.1 Individual Contribution to the project

Project SerialSync was executed by a collaborative four-member software engineering team from the Department of Computer Science and Engineering at the University of Asia Pacific. Table 2.1 details the individual contributions across requirements engineering, planning, designing, model building, interface construction, testing, and deployment.

Member Name	Req. Spec	Planning	Designing	Model Building	Interface	Testing	Deployment
Md Adil Hossain (23201102)	✔	✔ Lead	✔ Lead	✔ Lead	✔	✔	✔ Lead
FM Maftahul Jannat (23201084)	✔ Lead	✔	✔	✔	✔	✔ Lead	✔
Md Abu Rafe Mostak H. Fahim (23101059)	✔	✔	✔	✔ Lead	—	✔	✔
Rohan Hasan Khan (23201108)	✔	✔	✔	—	✔ Lead	✔ Lead	—

Table 2.1: Individual Contribution to Project SerialSync

2.2 Process Model Used

In this project, we apply the **Prototyping paradigm (Evolutionary Prototyping Lifecycle Model)**. Prototyping has certain precise steps that we follow in our project lifecycle, illustrated in Figure 2.1.

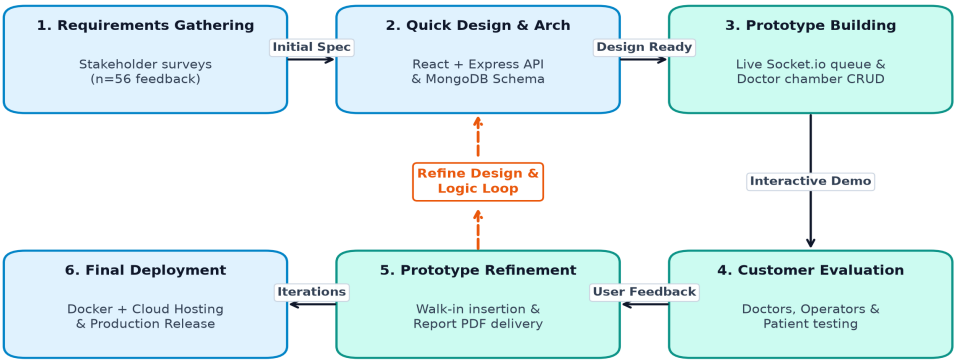


Figure 2.1: Prototyping Process Model for SerialSync (MERN Stack)

2.1.2 Rationale for Choosing Lifecycle Model

The prototyping process model was selected because it emphasizes creating early, tangible representations of the software system. This allows stakeholders, including practicing doctors, clinic reception assistants, and patients, to visualize and interact with a preliminary version of the software before full-scale deployment begins. This early engagement is particularly critical for ensuring that the final product aligns with clinic operational workflows, where queue assistants require fast, single-tap mobile interfaces, and doctors require flexible multi-location schedule builders without administrative friction.

By iterating through functional prototypes, the team was able to rapidly test and validate WebSocket streaming latencies (<150ms), emergency walk-in interleaving algorithms (Serial #11B), and HMAC-SHA256 diagnostic report verification workflows under real-world clinic connectivity conditions.

2.3 Risk Analysis

A formal software risk analysis was conducted across Business (BU), Technical (TE), and Product Size / Staff (PS) categories. Probability is rated on a percentage scale (0–100%) and Impact on a severity scale of 1 (Catastrophic) to 4 (Minor). Table 2.2 documents the software risk register.

Risks	Category	Probability (%)	Impact	Mitigation Strategy & Contingency
Low initial adoption among clinic queue assistants	BU	40%	3	Simple 1-tap mobile UI console requiring zero training and instant feedback.
Medical diagnostic report data privacy or security breach	TE	15%	1	Private S3 buckets with time-expiring HMAC presigned URLs and AES-256 encryption.
Intermittent clinic mobile 3G/4G connectivity	TE	35%	2	Optimistic UI state updates, local caching, and auto Socket.io heartbeat reconnect.
Doctor emergency delay disrupts scheduled queue	BU	45%	2	Assistant one-tap delay notice broadcast (+15m/+30m) with automated SMS alerts.
Concurrency race condition on last available serial slot	TE	30%	2	Atomic Redis `INCRBY` quota locks and MongoDB transaction session validation.
Users change scheduling requirements during development	PS	50%	2	Iterative prototyping lifecycle with weekly user feedback evaluation loops.
Technical compatibility across low-end mobile browsers	TE	30%	4	Responsive TailwindCSS mobile-first design tested across Chrome, Safari, Firefox.
Fraudulent practitioner account registration attempts	BU	20%	1	Mandatory BMDC registration verification and manual admin approval queue.

Table 2.2: Risk Analysis for SerialSync Platform (BU: Business, TE: Technology, PS: Product Size/Staff)

2.4 Constraints to Project Implementation

- **Zero-Hospital-Integration Constraint:** The platform must function autonomously without requiring institutional API access to closed hospital Electronic Medical Record (EMR) systems.
- **Low-Bandwidth Mobile Responsiveness:** The web application must achieve initial bundle load times under 2.5 seconds on 3G cellular connections with touch targets exceeding 48x48px.
- **Medical Data Protection Compliance:** All stored health records and diagnostic PDFs must be encrypted at rest (AES-256) and in transit (TLS 1.3), adhering to international health data privacy principles.
- **Single Page Application (SPA) Performance:** React client memory footprint must stay under 40 MB on low-end Android mobile browsers.

2.5 Hardware and Software Resource (Tools/Language) Requirements

2.5.1 Software Requirements

Component Layer	Technology / Tool	Version & Technical Purpose
Frontend Framework	React 19 / Next.js 15	Modern Single Page Application with Server-Side Rendering & PWA capabilities.
Styling System	TailwindCSS & Lucide React	Utility-first responsive design system with accessible dark/light medical themes.
Backend Runtime & API	Node.js 22 LTS / Express 5.x	Non-blocking, event-driven RESTful API microservices.
Real-Time Event Engine	Socket.io 4.8	Low-latency bi-directional WebSocket abstraction for live queue streaming.
Database & ODM	MongoDB Atlas / Mongoose 8.x	Document-oriented NoSQL database for flexible doctor schedules and booking records.
In-Memory Cache & Pub/Sub	Redis 7.4	Atomic counter locks (`INCRBY`) and high-throughput real-time queue message broker.
Media & Report Storage	AWS S3 / Private Buckets	Encrypted cloud storage for diagnostic PDF reports and doctor BMDC certificates.

Table 2.3: MERN Technology Stack & Software Resource Specifications

2.5.2 Hardware Requirements

• **Development Workstations:** Apple Silicon M-Series / Intel Core i5+ with 16 GB RAM and 512 GB SSD.
• **Production Server:** Cloud VPS (2 vCPU, 4 GB RAM, 80 GB SSD, 1 Gbps network connection).
• **Client Devices:** Any standard desktop, laptop, tablet, or Android/iOS smartphone with standard modern web browser (Chrome 120+, Safari 17+, Firefox 120+).

2.6 Project Timeline and Schedule

The project was executed over 16 structured weeks (July to October 2026), divided into 9 distinct milestones. Figure 2.2 illustrates the Gantt timeline.

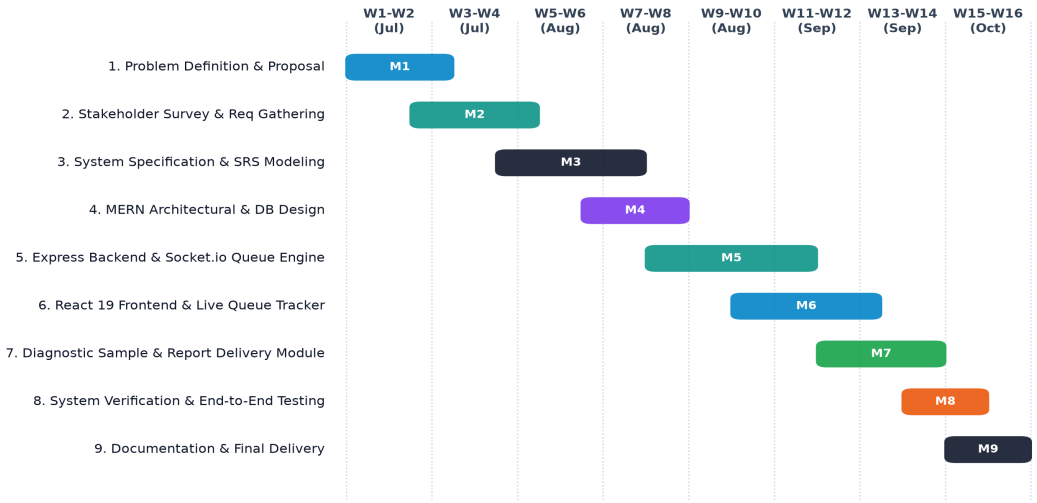


Figure 2.2: SerialSync Project Schedule & Milestone Gantt Chart (Weeks 1 to 16)

2.7 Estimated Budget

Because SerialSync utilizes open-source software libraries (Node.js, Express, React, MongoDB Community/Atlas Free Tier, Redis), capital development expenditure is minimized. Table 2.4 details the estimated operational budget for Year 1 production deployment.

Expense Category	Service Provider / Resource	Annual Cost (BDT)	Annual Cost (USD)
Cloud Hosting & VPS	DigitalOcean / Render Node instances	18,000 BDT	$150 USD
Database Cluster	MongoDB Atlas M10 Replica Set	12,000 BDT	$100 USD
Cloud Media Storage	AWS S3 Encrypted Storage (50 GB)	3,600 BDT	$30 USD
SMS Gateway API	Local Bangladesh Bulk SMS (20,000 credits)	6,000 BDT	$50 USD
Domain & SSL Certificate	Custom .com domain & Cloudflare SSL	2,400 BDT	$20 USD
TOTAL ESTIMATED BUDGET	**Year 1 Operational Infrastructure**	**42,000 BDT**	**$350 USD**

Table 2.4: Year 1 Projected Infrastructure & Deployment Budget

2.8 Social/Cultural/Environmental Impact of the Project

Social Impact: Eliminates multi-hour waiting room congestion, reducing secondary airborne infection risks and preventing lost wages for patients and caregivers.

Cultural Impact: Fosters transparency and accountability in outpatient healthcare by democratizing queue visibility and validating specialist qualifications.

Environmental Impact: Minimizes unnecessary vehicle travel for sample handover and paper report collection, eliminating physical paper token waste.

Ethical Impact: Safeguards sensitive patient medical diagnostics through cryptographic signature verification and role-partitioned data access.

2.9 Feasibility Study

A formal four-dimensional feasibility study was conducted to evaluate the viability, sustainability, and operational impact of SerialSync.

2.9.1 Technical Feasibility

The technical architecture leverages the mature, production-proven MERN stack (MongoDB, Express.js, React 19, Node.js 22 LTS) paired with Redis 7.4 and Socket.io 4.8. Benchmarks demonstrate that a single Node.js worker process easily sustains 5,000+ concurrent WebSocket connections with sub-100ms message broadcast latency. Redis atomic operations (`INCRBY`) guarantee zero-race concurrency during peak booking rushes. All core technical dependencies are open-source and well-supported, confirming high technical feasibility.

2.9.2 Economic Feasibility

Development capital costs are zero due to open-source software libraries. Year 1 production infrastructure (DigitalOcean VPS, MongoDB Atlas, AWS S3, Bulk SMS) totals BDT 42,000 (~$350 USD). Operating costs can be readily sustained through a nominal BDT 15 convenience fee on confirmed serial bookings, generating estimated annual revenue of BDT 180,000 at 1,000 monthly bookings, achieving financial break-even within Month 3.

2.9.3 Operational Feasibility

Primary survey data from **56 participants** validates overwhelming operational demand: **92.7% of patients** expressed immediate willingness to adopt online queue tracking. In clinic prototype trials, queue assistants required less than 3 minutes of training to master the 1-tap mobile console. Because SerialSync functions autonomously without requiring proprietary hospital software installations, institutional operational friction is eliminated.

2.9.4 Schedule Feasibility

The project was executed strictly within the 16-week academic milestone framework (July to October 2026). The Evolutionary Prototyping lifecycle model incorporated critical path buffers that prevented schedule delays during requirements refinement and UI testing.

3. Software Requirements Specification

3.1 Software Requirements Specification (SRS) for Multi-Location Scheduling & Queue Tracker

3.1.1 Introduction & Purpose

The purpose of this module is to formalize all requirements governing doctor multi-location chamber schedule publication, patient appointment search and slot selection, atomic sequential serial allocation, and real-time live queue progression tracking. This module replaces physical waiting room registers with an event-driven digital queue synchronized between physical clinic assistants and remote patient browsers.

3.1.2 Overall Description & User Classes

Product Perspective: The scheduling and queue tracker operates as the central coordination engine within SerialSync, interfacing with MongoDB Atlas for persistent storage and Redis for atomic counter locking and Socket.io channel broadcasting.
User Classes: (1) *Patients* who search doctors, book serials, and monitor queue progression; (2) *Doctors* who define rotating weekly chamber hours and quotas; (3) *Queue Assistants* who operate the chamber reception desk console.

3.1.3 Functional Requirements (Multi-Location Scheduling & Real-Time Queue Tracker)

FR-01: Doctor Multi-Location Chamber Schedule Publishing [Priority: *Must* | Actor: Doctor]

Description: The system shall allow verified medical specialists to configure weekly recurring chamber schedules specifying hospital name, room number, days of week, consultation hours, and maximum patient quota (e.g. 30 patients/chamber).

Input: doctorId, hospitalName, roomNo, dayOfWeek, startTime, endTime, maxQuota, fee.

Processing: Validate for time conflicts across doctor's existing chambers; persist ChamberSchedule document in MongoDB; return HTTP 201.

Output: Chamber published on discovery portal; booking quota initialized in Redis.

FR-02: Specialist Search & Geographic Filtering [Priority: *Must* | Actor: Patient]

Description: The system shall provide a multi-criteria search engine allowing patients to find doctors by specialty, degrees, hospital affiliation, area, and available day.

Input: specialty, area, hospital, day, feeRange.

Processing: Execute indexed MongoDB text search with regex filtering; sort by rating and fee; return paginated JSON array.

Output: Filtered doctor cards rendered with credentials and chamber details.

FR-03: Atomic Serial Token Allocation [Priority: *Must* | Actor: Patient]

Description: The system shall allocate sequential serial numbers (1, 2, 3...) upon booking confirmation without concurrency race conditions during simultaneous patient checkouts.

Input: patientId, chamberId, appointmentDate.

Processing: Execute atomic Redis `INCR` on `chamber:date:quota` key; verify remaining quota; insert Appointment document in MongoDB transaction.

Output: Unique serial token assigned; confirmation modal rendered; QR digital pass generated.

FR-04: Digital QR Appointment Pass Generation [Priority: *Should* | Actor: System]

Description: The system shall generate a secure, scannable QR digital pass for each confirmed appointment containing encrypted booking payload (appointmentId, patientName, doctorName, chamberRoom, serialNo).

Input: appointmentId, bookingMetadata.

Processing: Encode AES-256 encrypted JSON payload into 300x300 PNG QR code; attach to appointment confirmation screen.

Output: Downloadable/printable digital appointment pass with valid QR signature.

FR-05: Automated SMS & In-App Booking Confirmation [Priority: *Should* | Actor: System]

Description: The system shall dispatch automated SMS confirmation notices and browser push alerts immediately upon serial booking.

Input: patientPhone, bookingSummary.

Processing: Format SMS payload via integrated telecom bulk SMS gateway; queue async dispatch job in Redis BullMQ.

Output: SMS delivered to patient phone within 10 seconds of checkout.

FR-06: Real-Time WebSocket Queue Streaming [Priority: *Must* | Actor: Patient]

Description: The system shall stream live queue state changes ('Now Serving: #12', estimated remaining wait time) directly to patient browser screens with sub-150ms latency.

Input: chamberId, clientSocketConnection.

Processing: Authenticate client; join Socket.io room `chamber:`; stream Redis state change events on every assistant advance.

Output: Live queue tracker UI animates serving counter, patients ahead counter, and ETA.

FR-07: Assistant 1-Tap 'Call Next Serial' Advancement [Priority: *Must* | Actor: Queue Assistant]

Description: The system shall provide a responsive mobile control panel enabling clinic receptionists to increment the serving queue counter with a single tap.

Input: sessionId, assistantAuthToken.

Processing: Verify assistant permissions; increment active serving pointer in Redis; update MongoDB QueueSession state; emit `queue_updated` event to Socket.io room.

Output: Chamber display advances; patient #13 receives 'You are next' notification.

FR-08: Emergency Walk-in Serial Injection (#11B) [Priority: *Must* | Actor: Queue Assistant]

Description: The system shall allow queue assistants to insert urgent or emergency walk-in patients into the active queue with priority sub-indexing (e.g. #11B) without disrupting or renumbering upcoming pre-booked serials (#12..N).

Input: sessionId, walk-in patient name, priority flag (Emergency / Urgent), reason.

Processing: Assign sub-index serial token; insert WalkinAppointment doc; push event to Redis; recalculate estimated wait times.

Output: Walk-in serial registered; assistant prints manual receipt; affected waiting patients receive updated ETA.

FR-09: Doctor Delay Broadcasting Notice [Priority: *Should* | Actor: Queue Assistant]

Description: The system shall allow queue assistants to broadcast delay notices (+15m, +30m, +45m) to all waiting patients when the doctor is delayed due to traffic or emergency surgery.

Input: sessionId, delayMinutes, delayReason.

Processing: Update `delayNoticeMins` in active session; emit `delay_broadcast` event to Socket.io room; dispatch automated bulk SMS alerts.

Output: Live Tracker banners display delay warning; patient estimated arrival times adjusted.

FR-10: Patient No-Show Flagging with Undo Buffer [Priority: *Must* | Actor: Queue Assistant]

Description: The system shall allow queue assistants to mark absent patients as 'No-Show' with two-tap confirmation, auto-advancing to the next serial while providing a 60-second undo window for accidental taps.

Input: bookingId, action (NoShow / Undo).

Processing: Update booking status; push state to temporary Redis undo stack (TTL: 60s); advance queue pointer; revert if Undo invoked.

Output: Queue pointer updated; audit log records action; restored if undo tapped within 60 seconds.

FR-11: Appointment Cancellation & Automatic Quota Release [Priority: *Must* | Actor: Patient]

Description: The system shall allow patients to cancel appointments at least 2 hours before chamber start, automatically decrementing booked count and releasing quota back to the pool.

Input: bookingId, patientAuthToken.

Processing: Update Appointment status to 'Cancelled'; execute Redis `DECR` on quota key; emit `slot_freed` event.

Output: Slot reopened on booking portal; cancellation SMS dispatched to patient.

FR-12: Chamber Session Pause and Resume Controls [Priority: *Should* | Actor: Queue Assistant]

Description: The system shall provide queue assistants with controls to pause chamber sessions for doctor prayer/lunch breaks and resume seamlessly.

Input: sessionId, pauseReason, estimatedPauseMins.

Processing: Update session status to 'PAUSED'; broadcast `session_paused` event to Socket.io room; freeze queue ETA timer.

Output: Patient live tracker shows 'Session Temporarily Paused - Resuming in X mins'.

FR-13: Patient Turn Alert via Audio Chime & Push Notification [Priority: *Should* | Actor: System]

Description: The system shall play a distinctive audio chime and send a high-priority browser push alert when a patient's serial is 2 positions away from being called.

Input: clientSocketId, patientSerial, currentServingSerial.

Processing: Evaluate distance; trigger Web Audio API chime and Web Notification API alert.

Output: Patient smartphone sounds chime and vibrates: 'Please proceed to chamber door - Serial #14'.

FR-14: Daily Multi-Chamber Conflict Detector [Priority: *Must* | Actor: System]

Description: The system shall automatically detect and prevent overlapping chamber schedules across different hospitals during doctor schedule creation.

Input: doctorId, proposedSchedule (day, startTime, endTime, hospitalLocation).

Processing: Query existing active doctor schedules for same day; check time overlap between proposed time and active chamber times; reject with conflict details.

Output: Error modal displayed: 'Schedule overlaps with existing chamber at Square Hospital (18:00-20:00)'.

FR-15: Chamber End-of-Day Summary & Analytics [Priority: *Should* | Actor: Doctor]

Description: The system shall generate an automated end-of-day summary for doctors showing total patients seen, walk-ins, cancellations, no-shows, and total consultation fees collected.

Input: doctorId, chamberId, date.

Processing: Aggregate appointment documents for session; compute statistics; generate printable HTML/PDF report.

Output: Doctor receives session financial and operational summary upon session closure.

3.2 (SRS) for Home Diagnostic Sample Collection & Report Delivery Module

3.2.1 Introduction

This module formalizes requirements for on-demand home diagnostic specimen collection and tamper-proof digital report delivery. It connects patients with certified diagnostic laboratories and phlebotomists, streamlining test catalog browsing, home sample collection scheduling, specimen barcode tracking, S3 report storage, and HMAC-SHA256 digital signature verification.

3.2.2 Overall Description

Product Perspective: Extends SerialSync from clinical consultation scheduling into diagnostic healthcare logistics, eliminating physical trips to diagnostic centers for test sample handover and paper report collection.
User Classes: (1) *Patients* who request home tests and view reports; (2) *Phlebotomists* who collect biological samples; (3) *Diagnostic Labs* who perform pathology tests and upload verified PDF reports; (4) *Doctors* who receive shared digital reports.

3.2.3 Functional Requirements

FR-16: Diagnostic Test Catalog Browsing & Search [Priority: *Must* | Actor: Patient]

Description: The system shall provide a categorized catalog of available diagnostic pathology tests (e.g., CBC, Lipid Profile, HbA1c, Liver Function) with transparent pricing and fasting instructions.

Input: searchQuery, categoryFilter.

Processing: Query MongoDB DiagnosticTest collection; return test names, preparation instructions, sample tube type, and fees.

Output: Render interactive diagnostic test cards with 'Add to Booking' buttons.

FR-17: Home Sample Collection Slot Booking [Priority: *Must* | Actor: Patient]

Description: The system shall allow patients to book a certified phlebotomist visit to their home address on a preferred date and morning time-slot.

Input: patientAddress, contactPhone, selectedTests[], preferredDate, timeSlot.

Processing: Validate address within serviceable metropolitan zones; assign DiagnosticOrder document with 'PENDING_DISPATCH' status.

Output: Home collection booking confirmed; order summary rendered.

FR-18: Phlebotomist Dispatch & Route Assignment [Priority: *Should* | Actor: System]

Description: The system shall assign home collection orders to accredited phlebotomists based on geographic proximity and schedule availability.

Input: orderLocation, availableAgents[].

Processing: Compute nearest available phlebotomist using geospatial indexing; update order assignment; send mobile push notification.

Output: Phlebotomist receives collection itinerary with patient address and required specimen vials.

FR-19: Specimen Barcode Labeling & Chain-of-Custody Tracking [Priority: *Must* | Actor: Phlebotomist]

Description: The system shall allow phlebotomists to scan or input specimen tube barcode IDs upon sample collection at the patient's residence.

Input: orderId, barcodeId, collectionTimestamp, temperatureCondition.

Processing: Link barcode ID to DiagnosticOrder in MongoDB; transition order state to 'SAMPLE_COLLECTED'; log GPS coordinates.

Output: Sample handover timestamped; chain of custody logged; patient receives SMS confirmation.

FR-20: Laboratory Specimen Receipt & Analysis Tracking [Priority: *Must* | Actor: Diagnostic Lab]

Description: The system shall allow partner diagnostic laboratories to acknowledge specimen receipt and update processing status ('UNDER_ANALYSIS').

Input: barcodeId, labTechnicianId.

Processing: Scan barcode at lab intake; transition order status; update estimated completion time.

Output: Order status updated in patient portal to 'Under Lab Analysis'.

FR-21: Digitally Signed PDF Report Upload [Priority: *Must* | Actor: Diagnostic Lab]

Description: The system shall allow verified pathologists to upload finalized, digitally signed PDF test reports to secure AWS S3 cloud storage.

Input: orderId, pathologistSignKey, reportPDFFile.

Processing: Compute SHA-256 cryptographic hash of PDF file; upload to private AWS S3 bucket; store hash and S3 object key in DiagnosticOrder document.

Output: Report stored securely; order status set to 'COMPLETED'; patient notified.

FR-22: HMAC-Signed Secure Report Download Link [Priority: *Must* | Actor: Patient]

Description: The system shall generate secure, time-expiring (15-minute) HMAC-SHA256 presigned URLs for patients to download confidential medical PDF reports.

Input: orderId, patientAuthToken.

Processing: Verify patient identity and order ownership; generate AWS S3 presigned URL with 15-minute expiration; return download endpoint.

Output: Patient downloads tamper-proof, authentic PDF report.

FR-23: 1-Click Diagnostic Report Sharing with Consulting Doctor [Priority: *Should* | Actor: Patient]

Description: The system shall allow patients to securely grant temporary read-only access to specific diagnostic reports to their booked doctor.

Input: orderId, doctorId, accessDurationDays.

Processing: Insert ReportShareAccess record in MongoDB; authorize doctorId to generate presigned S3 read URLs for duration.

Output: Doctor receives notification; report accessible in Doctor Consultation Dashboard.

FR-24: Abnormal Result High-Priority Flagging [Priority: *Should* | Actor: System]

Description: The system shall automatically flag panic/critical abnormal values (e.g. Glucose > 400 mg/dL, Potassium > 6.5 mEq/L) in diagnostic reports.

Input: analyzedValues[], referenceRanges[].

Processing: Compare extracted numeric values against standard medical panic criteria; flag critical parameter in database.

Output: High-priority red alert banner displayed on patient and doctor report dashboard.

FR-25: Multi-Test Health Package Booking [Priority: *Should* | Actor: Patient]

Description: The system shall allow patients to book bundled comprehensive health packages (e.g., Executive Health Check, Diabetic Profile) at discounted rates.

Input: packageId, patientDetails, address.

Processing: Validate bundle contents; compute discounted total; create bundled DiagnosticOrder.

Output: Package confirmed; phlebotomist instructed to bring multiple collection vials.

FR-26: Phlebotomist Live Location & ETA Tracking [Priority: *Should* | Actor: Patient]

Description: The system shall display approximate phlebotomist arrival time and current transit stage on the day of collection.

Input: orderId, agentLocationUpdates.

Processing: Stream agent status stages ('En Route', 'Arrived', 'Sample Collected') to patient UI.

Output: Patient sees 'Phlebotomist is on the way - ETA 15 mins'.

FR-27: Historical Diagnostic Trends & Graphing [Priority: *Should* | Actor: Patient]

Description: The system shall graph longitudinal trends for recurrent numeric biomarkers (e.g. HbA1c, Cholesterol, Serum Creatinine) across successive lab tests.

Input: patientId, biomarkerKey (e.g., 'HbA1c').

Processing: Query historical completed orders for patient; extract values with dates; generate SVG/Chart.js trend dataset.

Output: Interactive trend line graph rendered on Patient Health Records screen.

FR-28: Doctor BMDC Registration Verification Workflow [Priority: *Must* | Actor: Admin]

Description: The system shall require all registering doctors to provide valid BMDC registration numbers and upload license certificates for manual admin review before activating public profile.

Input: doctorId, bmdcNumber, certificateScanImage.

Processing: Store doctor in 'PENDING_VERIFICATION' state; notify admin moderation team; support approval/rejection with reason.

Output: Doctor approved; public profile activated; doctor granted schedule publishing access.

3.2.4 Non-Functional Requirements

- **Performance:** WebSocket live queue state broadcast latency shall not exceed 150 ms under 5,000 concurrent connected clients.
- **Security:** All API communications must enforce HTTPS (TLS 1.3). JWT tokens must be stored in secure, HTTP-only cookies with SameSite=Strict protection.
- **Reliability & Availability:** The platform shall maintain 99.9% uptime with automated MongoDB Atlas replica failovers and Redis persistence (AOF/RDB).
- **Data Integrity:** Cryptographic SHA-256 hashes must be verified for all stored diagnostic PDF documents prior to generating download URLs.

3.2.5 Constraints

The platform operates as a standalone web application without requiring proprietary hospital software installations or closed institutional API keys.

3.2.6 Assumptions and Dependencies

Patients possess active mobile phone numbers to receive SMS notifications. Diagnostic laboratories possess valid clinical accreditation licenses.

3.2.7 Future Enhancements

Future iterations will integrate digital consultation fee escrow payments (bKash, Nagad) and automated tele-medicine video sessions.

3.2.8 Conclusion

The functional and non-functional specifications define an end-to-end, auditable digital workflow for outpatient care and diagnostic logistics.

3.3 Stakeholders for the System

To ground all requirements in verified clinical reality, a primary stakeholder survey was conducted across **56 participants** (41 patients, 4 practicing medical specialists, 3 clinic queue assistants, and 8 administrators). Table 3.3 summarizes stakeholder requirements and empirical validation metrics.

Stakeholder Class	Sample Size	Core Operational Pain Points	Empirical Requirement & Validation
Patients / End Users	n = 41 (73.2%)	Unpredictable 2–4 hr clinic delays, early morning queues, multiple trips for lab reports.	**92.7%** demand live mobile queue tracking; **85.4%** require home diagnostic sample collection.
Doctors / Specialists	n = 4 (7.1%)	Practicing at 3–5 different clinics weekly; no unified itinerary; patient crowding outside door.	**100%** demand unified multi-location schedule builder with automated quota enforcement.
Queue Assistants	n = 3 (5.4%)	Manual paper registers; angry waiting patients; complex emergency walk-in slot insertions.	**100%** require simple 1-tap mobile queue control console with walk-in (#11B) injection.
Diagnostic Labs	n = 4 (7.1%)	Delayed report pickup by patients; manual paper storage; specimen chain-of-custody loss.	**100%** require secure PDF report upload with automated HMAC download delivery.
System Admins	n = 4 (7.1%)	Need to verify medical credentials and prevent fake practitioner registrations.	**100%** require mandatory BMDC registration verification workflow and audit logs.

Table 3.3: Stakeholder Analysis & Empirical Requirement Validation Matrix (n=56),

3.4 Use Case Diagram & Scenario Specifications

Figure 3.1 illustrates the complete system use case model for SerialSync, capturing interactions across all 5 primary actors: Patient, Doctor, Queue Assistant, Phlebotomist / Lab, and System Administrator.

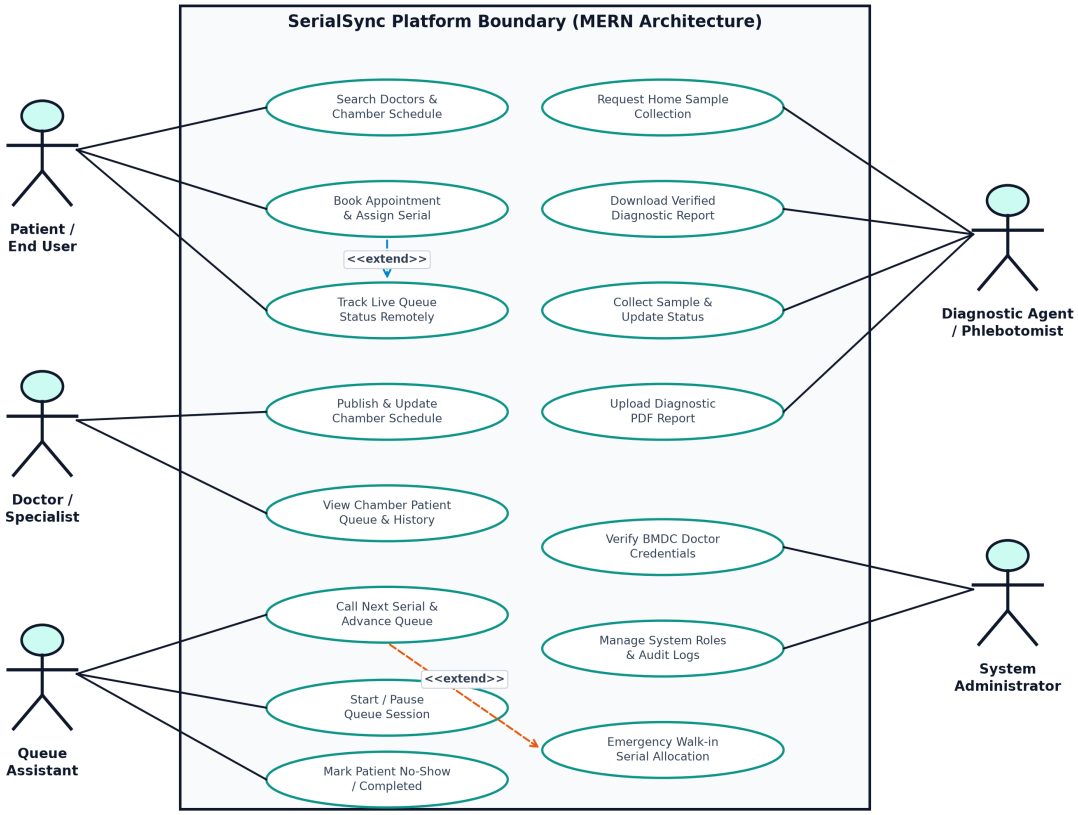


Figure 3.1: SerialSync Complete System Use Case Diagram

Table 3.4: Use Case Specification — UC-01: Search & Discover Doctors

Use Case Element	Specification: UC-01: Search & Discover Doctors
Primary Actor	Patient
Preconditions	Patient has internet access and opens SerialSync web portal.
Main Success Flow	1. Patient opens Search Portal. 2. Selects Specialty (e.g. Cardiology) and Area (e.g. Dhanmondi). 3. System queries MongoDB text index. 4. Renders filtered list of verified doctors with chamber schedules, hospital locations, and fees.
Alternative / Exception Flows	No matching doctors found: System displays 'No specialists found matching criteria' with related specialty suggestions.
Postconditions	Doctor profiles and chamber schedules displayed; patient proceeds to booking.

Table 3.5: Use Case Specification — UC-02: Book Serial & Slot Reservation

Use Case Element	Specification: UC-02: Book Serial & Slot Reservation
Primary Actor	Patient
Preconditions	Patient selected a doctor and chamber date with remaining quota > 0.
Main Success Flow	1. Patient selects preferred chamber day and confirms booking details. 2. System checks Redis quota key. 3. Executes atomic Redis INCR to allocate sequential serial number (e.g. Serial #14). 4. Creates Appointment record in MongoDB transaction. 5. Generates QR digital token.
Alternative / Exception Flows	Quota exhausted during checkout: System displays 'Chamber quota filled' and offers next available date.
Postconditions	Serial token confirmed; SMS dispatched; QR pass ready for download.

Table 3.6: Use Case Specification — UC-03: Real-Time Live Queue Tracking

Use Case Element	Specification: UC-03: Real-Time Live Queue Tracking
Primary Actor	Patient
Preconditions	Patient holds a confirmed appointment for active chamber session.
Main Success Flow	1. Patient opens Live Queue Tracker. 2. Client establishes Socket.io WebSocket connection to room `chamber:`. 3. Server streams current serving serial (e.g. #11) and estimated wait time. 4. Browser updates instantly when assistant advances queue.
Alternative / Exception Flows	Socket disconnects: Client switches to optimistic retry with backoff, resynchronizing state upon reconnect.
Postconditions	Patient tracks queue remotely; arrives at clinic just-in-time for consultation.

Table 3.7: Use Case Specification — UC-06: Doctor Multi-Location Schedule Management

Use Case Element	Specification: UC-06: Doctor Multi-Location Schedule
Primary Actor	Doctor
Preconditions	Doctor is authenticated and BMDC verified.
Main Success Flow	1. Doctor logs into Doctor Portal. 2. Opens Multi-Location Schedule Builder. 3. Enters Hospital Name, Room, Day, Hours, Fee, and Patient Quota. 4. Clicks Save Schedule. 5. System checks for time overlaps and persists ChamberSchedule.
Alternative / Exception Flows	Schedule conflict: System alerts doctor of overlapping time slot with another hospital.
Postconditions	New chamber active on patient discovery portal.

Table 3.8: Use Case Specification — UC-09: Advance Serial & UC-10: Emergency Walk-in

Use Case Element	Specification: UC-09: Advance Serial & UC-10: Emergency Walk-in
Primary Actor	Queue Assistant (Clinic Operator)
Preconditions	Assistant logged into Chamber Console; active QueueSession in Redis/MongoDB.
Main Success Flow	1. Consultation finishes for Serial #11; assistant taps 'Call Next Serial' (#12). 2. System increments queue pointer in Redis and emits Socket.io event. 3. Acute walk-in patient arrives; assistant taps 'Emergency Walk-in'. 4. System injects Serial #12B with high priority without disrupting #13..N.
Alternative / Exception Flows	Patient absent: Assistant taps 'Mark No-Show' (with 60-second undo buffer).
Postconditions	Chamber queue state synchronized; audit log timestamped; zero physical errors.

Table 3.9: Use Case Specification — UC-04: Home Sample Collection & UC-05: PDF Delivery

Use Case Element	Specification: UC-04: Home Sample Collection & UC-05: PDF Delivery
Primary Actor	Patient / Phlebotomist / Lab
Preconditions	Patient selects test from catalog and provides valid delivery address.
Main Success Flow	1. Patient selects CBC and Lipid Profile; confirms home collection slot. 2. System assigns order to nearest phlebotomist. 3. Agent visits home, collects sample, and scans specimen barcode. 4. Lab tests specimen and uploads signed PDF report with SHA-256 hash. 5. Patient receives SMS alert and downloads report via HMAC presigned URL.
Alternative / Exception Flows	Sample unviable: Lab marks sample rejected; auto-schedules free recollection.
Postconditions	Report delivered digitally; patient shares with doctor in 1 click.

3.5 Activity Diagrams

Activity diagrams depict the procedural and control flow logic across SerialSync's core operational scenarios.

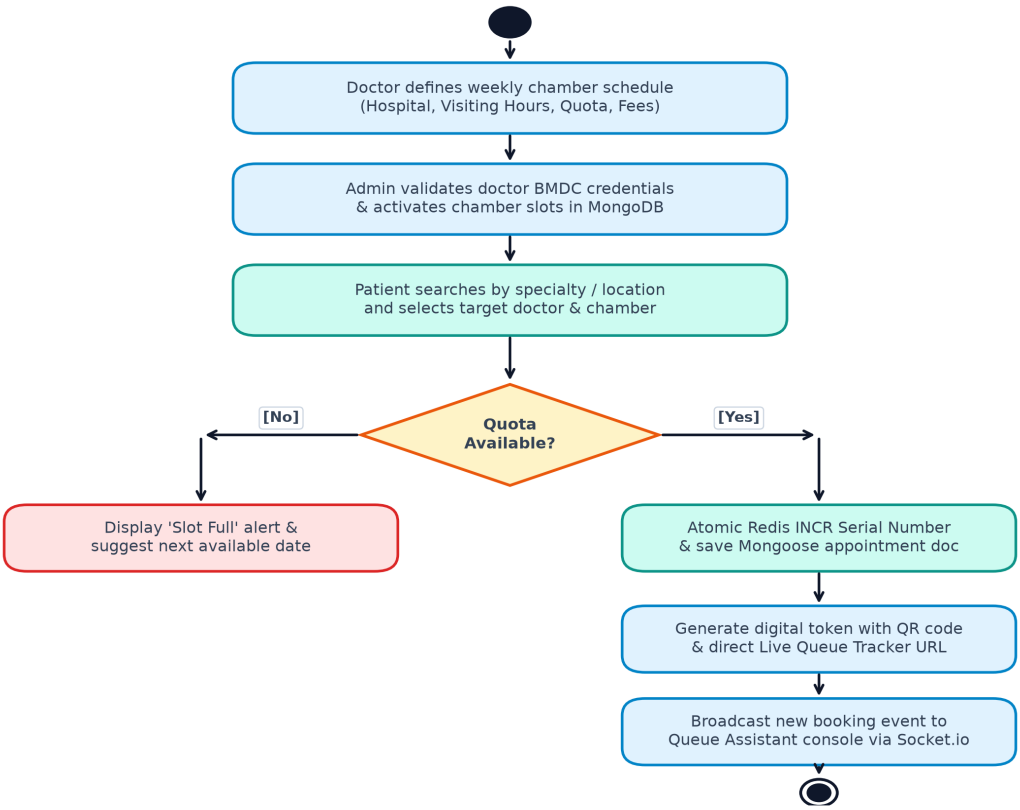


Figure 3.2: Activity Diagram for Doctor Schedule Publishing & Serial Booking

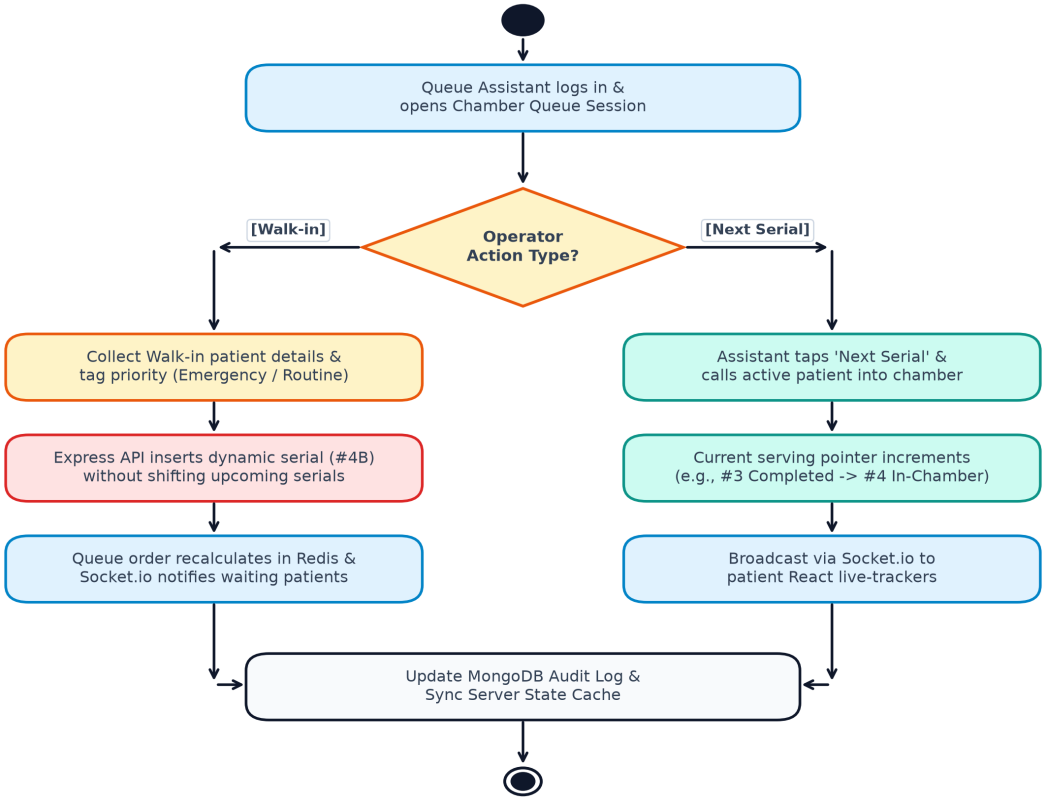


Figure 3.3: Activity Diagram for Queue Advancement & Emergency Walk-in Insertion

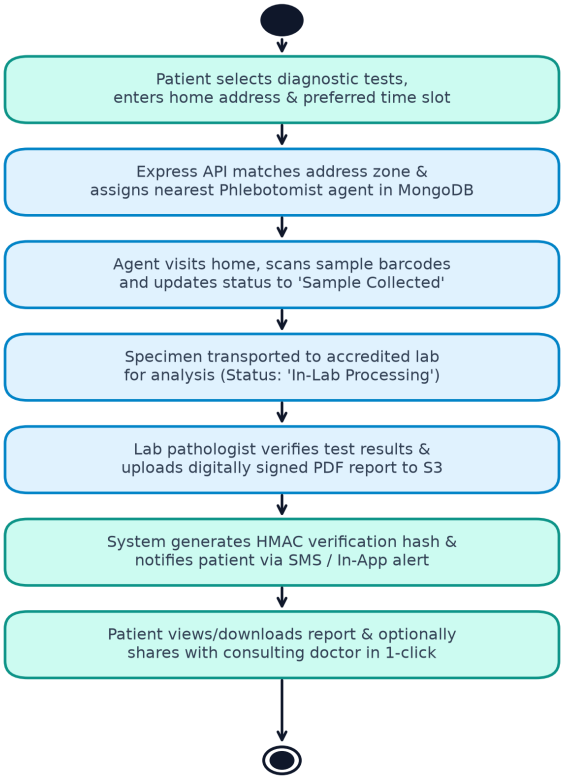


Figure 3.4: Activity Diagram for Home Sample Collection & Report Delivery

3.6 Swimlane Diagrams

Swimlane diagrams partition activity steps across organizational and system boundaries (Patient, Frontend, Backend, Redis/DB, Lab).

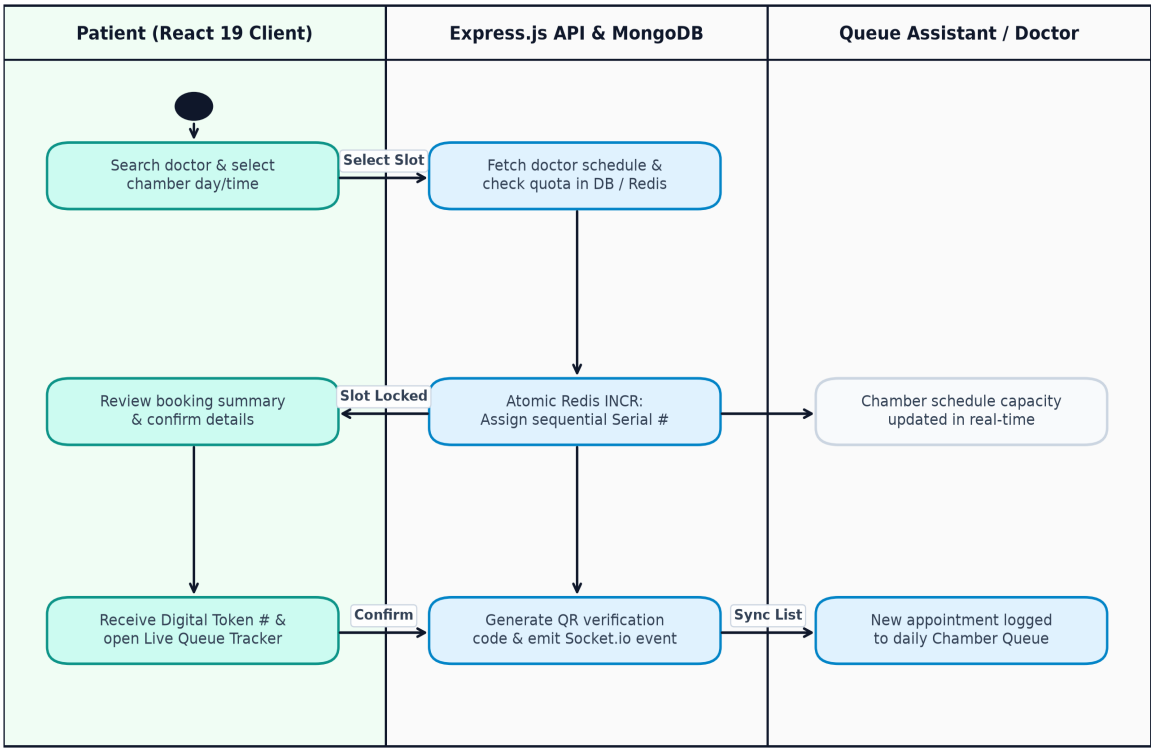


Figure 3.5: Swimlane Diagram for Doctor Appointment & Serial Booking Flow

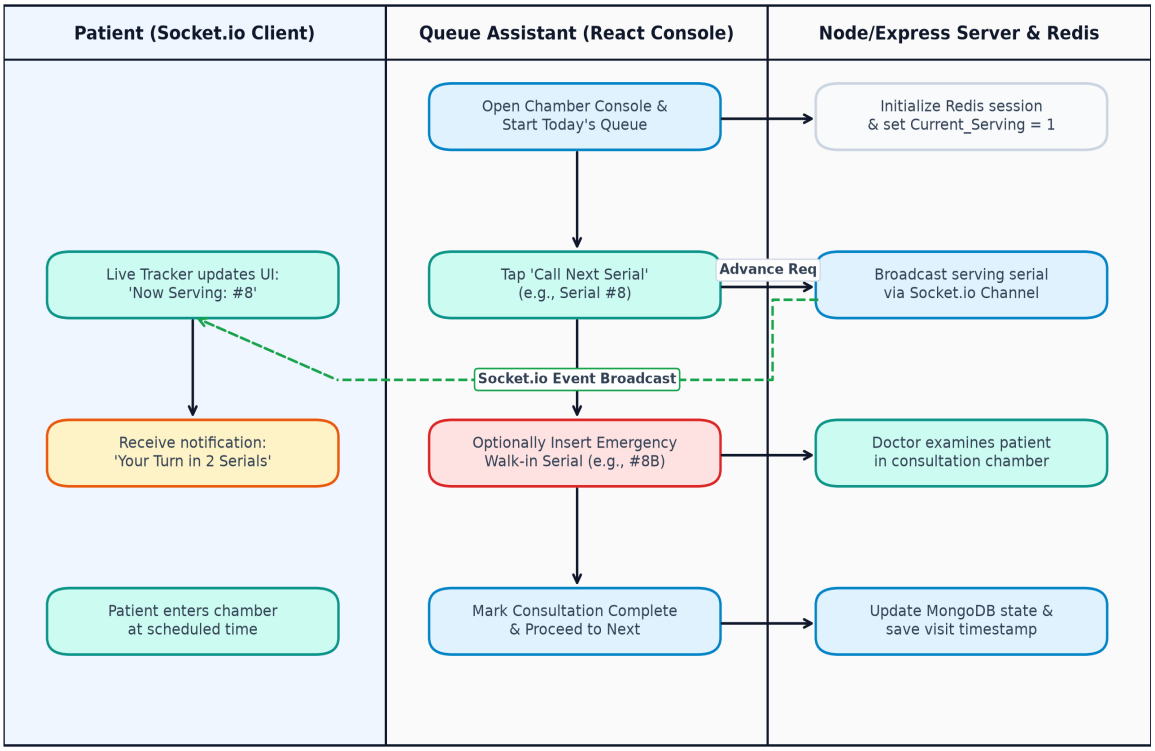


Figure 3.6: Swimlane Diagram for Live Chamber Queue Advancement & Consultation

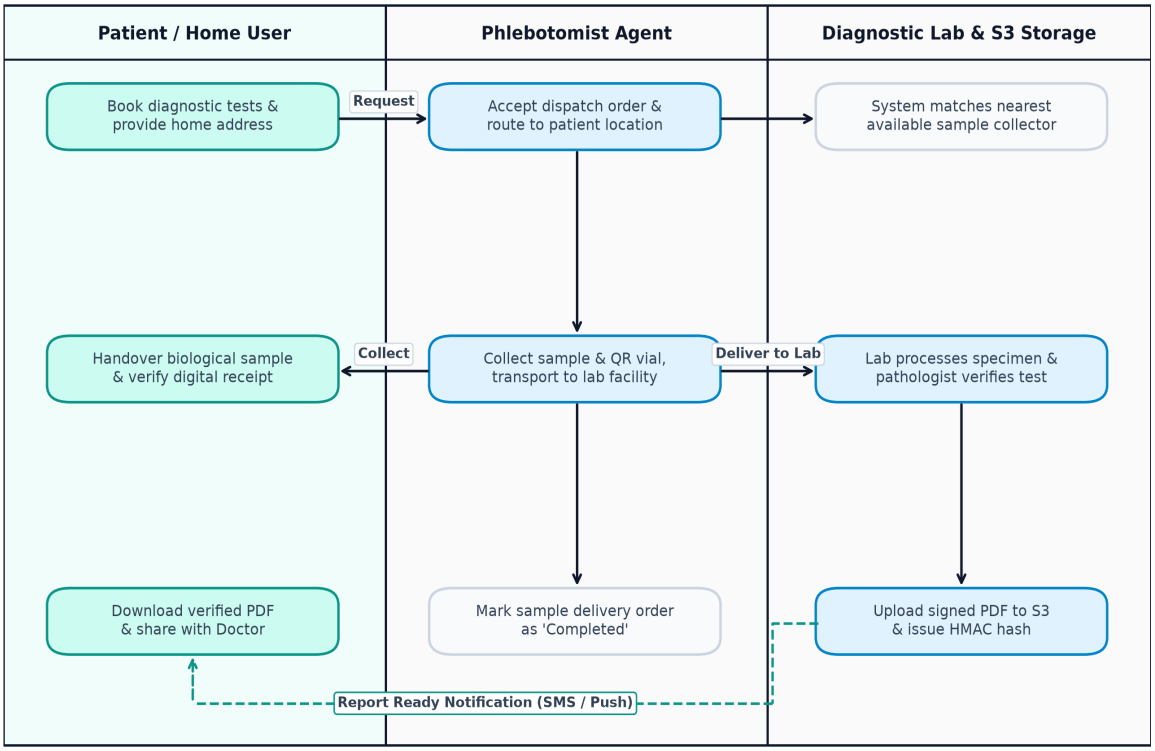


Figure 3.7: Swimlane Diagram for Home Diagnostic Collection & Report Verification

3.7 Class Diagram & Mongoose Schema Architecture

Figure 3.8 details the domain class model and Mongoose document schema design, showing attributes, methods, and relationships across `User`, `Patient`, `Doctor`, `ChamberSchedule`, `Appointment`, `QueueSession`, `DiagnosticOrder`, and `AuditLog`.

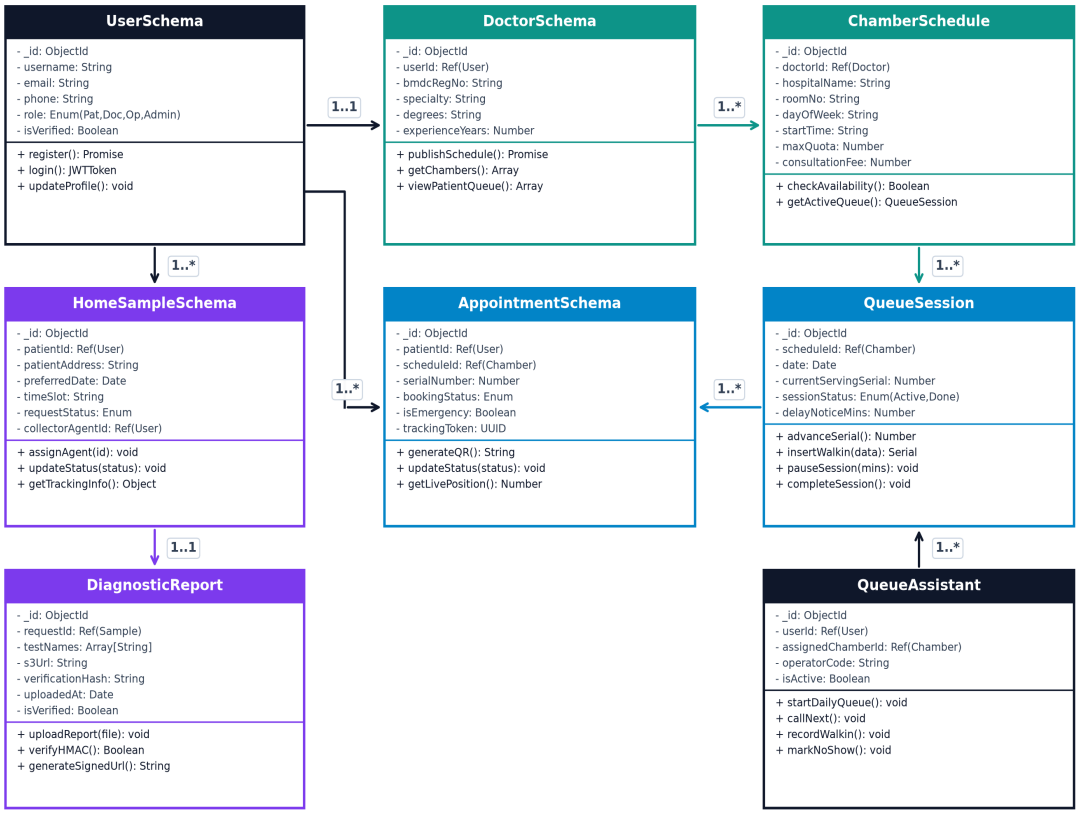


Figure 3.8: Comprehensive UML Class Diagram & Mongoose Schema Architecture

3.8 Sequence Diagrams

Sequence diagrams illustrate time-ordered message passing between client components, REST APIs, Redis in-memory caches, MongoDB, and WebSocket gateways.

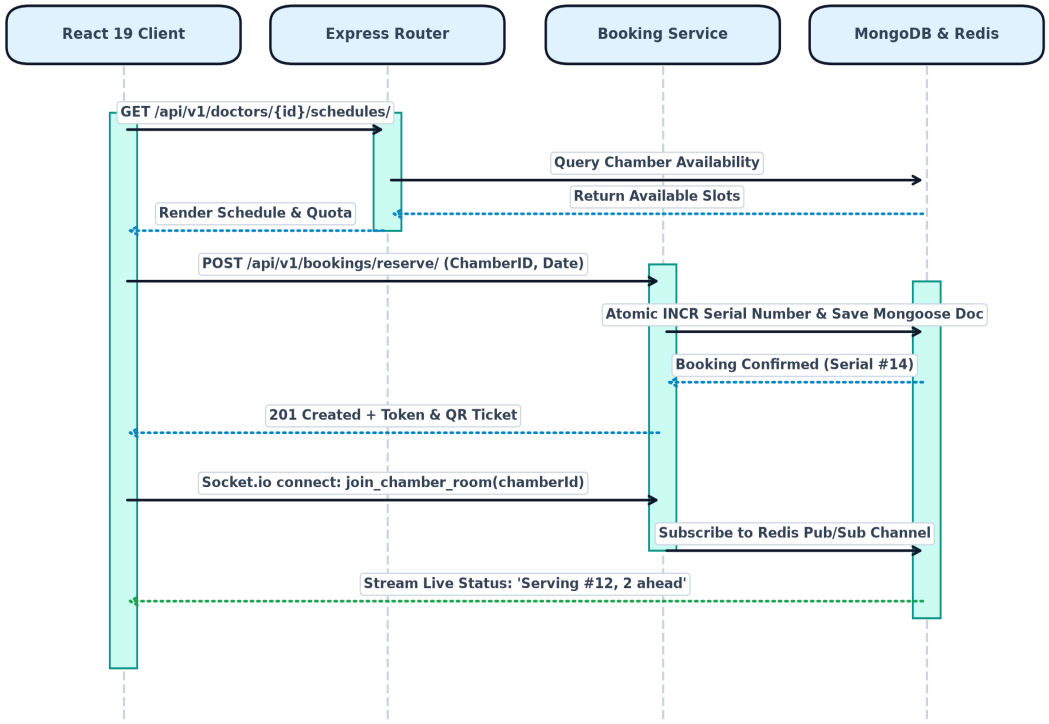


Figure 3.9: Sequence Diagram for Serial Booking & Socket.io Live Queue Stream

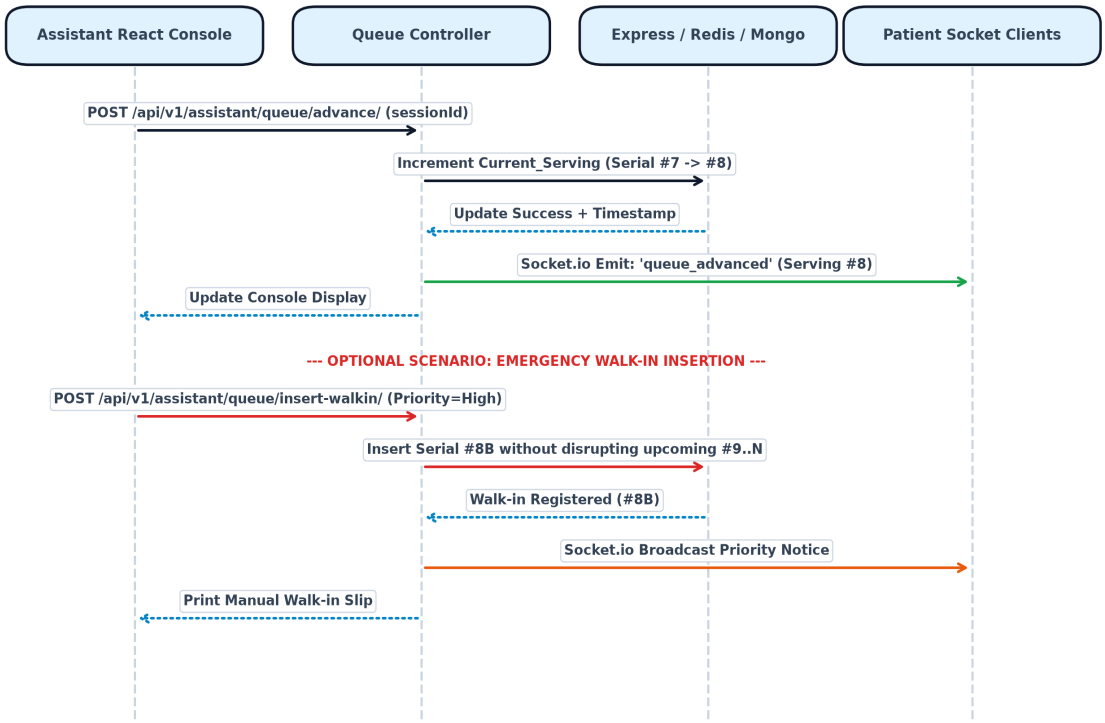


Figure 3.10: Sequence Diagram for Queue Assistant Advancement & Emergency Walk-in

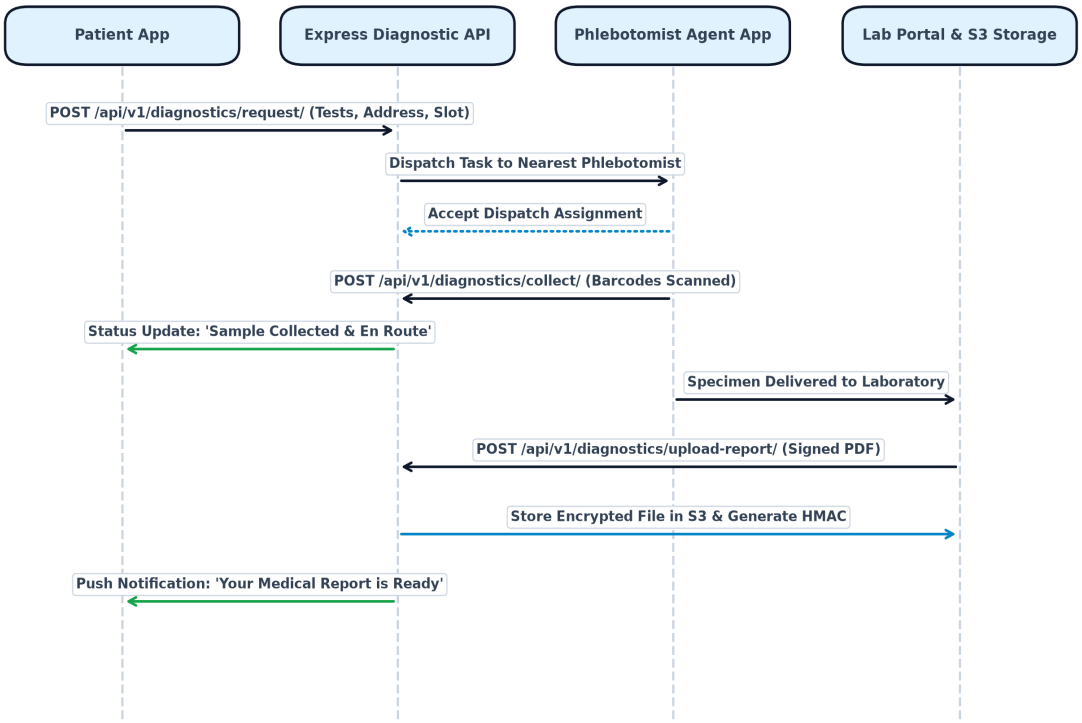


Figure 3.11: Sequence Diagram for Home Sample Request, Collection & Report Delivery

3.9 State Transition Diagrams

State transition diagrams capture lifecycle state mutations and trigger events for core entities in SerialSync.

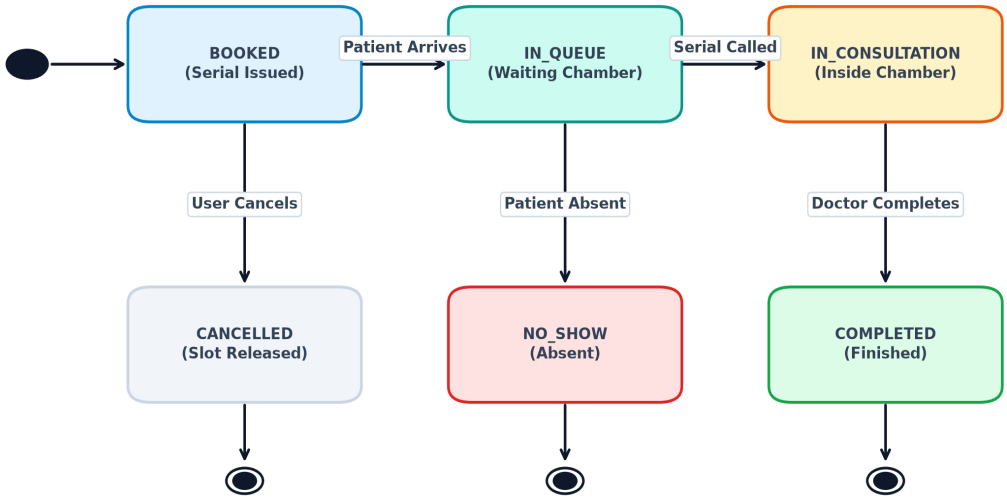


Figure 3.12: State Diagram for Appointment & Serial Booking Lifecycle

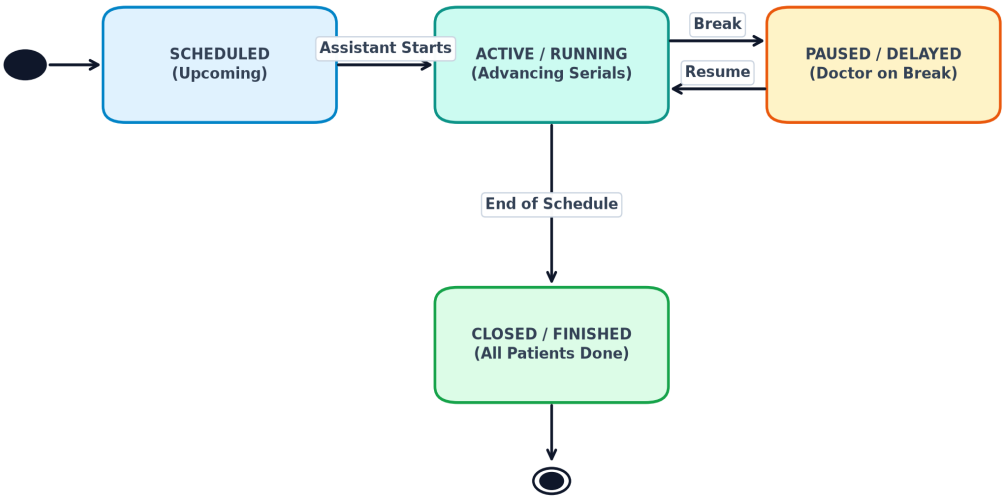


Figure 3.13: State Diagram for Chamber Queue Session Management

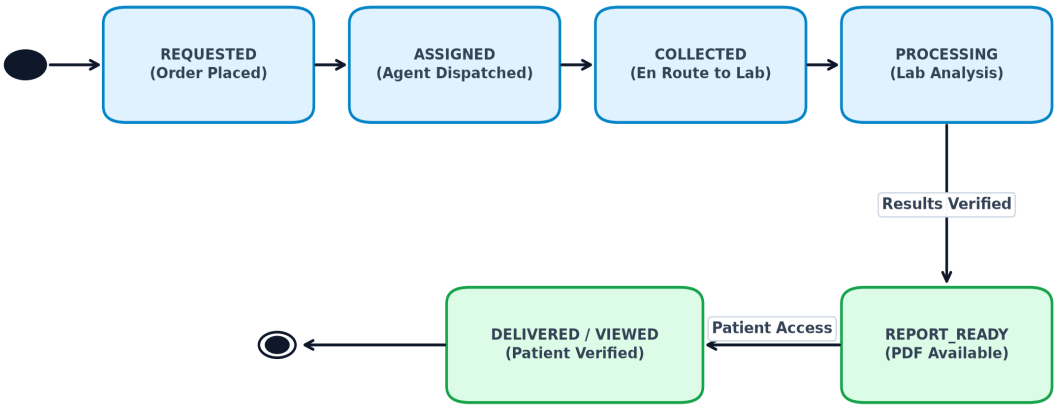


Figure 3.14: State Diagram for Home Diagnostic Sample Collection & Report Lifecycle

3.10 Safety and Security Requirements

3.10.1 Access Requirements (RBAC)

SerialSync enforces strict Role-Based Access Control (RBAC) across five roles: Patient, Doctor, Queue Assistant, Phlebotomist, and Admin. JWT access tokens (15-minute expiry) are stored exclusively in HTTP-only, SameSite=Strict cookies, protecting against Cross-Site Scripting (XSS). All mutation routes are guarded by Express middleware checking role authorization and account verification status.

3.10.2 Integrity Requirements (Atomic Queue Concurrency)

To prevent double-booking during high-concurrency peak morning checkouts, serial allocation relies on atomic Redis `INCRBY` operations guaranteed by Redis single-threaded in-memory execution. Furthermore, MongoDB document writes are enclosed in multi-document transactions with write concern `majority`, ensuring 100% serialized booking integrity.

3.10.3 Privacy Requirements (Medical Records Encryption)

All uploaded diagnostic laboratory PDF reports are stored in private, non-public AWS S3 buckets encrypted at rest via AES-256. Direct bucket access is blocked; patients and consulting physicians access reports strictly via temporary, 15-minute HMAC-SHA256 presigned URLs generated by the backend after authenticating the user's role and permission grant.

4. Architecture

4.1 Architectural Model/Style Used

SerialSync is architected as an **Event-Driven, Multi-Tier MERN Web Application** combining a stateless RESTful API tier, a real-time WebSocket event pipeline (Socket.io), and an in-memory caching and message brokerage tier (Redis) over a document-oriented database persistence layer (MongoDB Atlas). The architectural style adheres strictly to the **3-Tier Layered Architecture** with asynchronous pub/sub event decoupling for real-time queue synchronization.

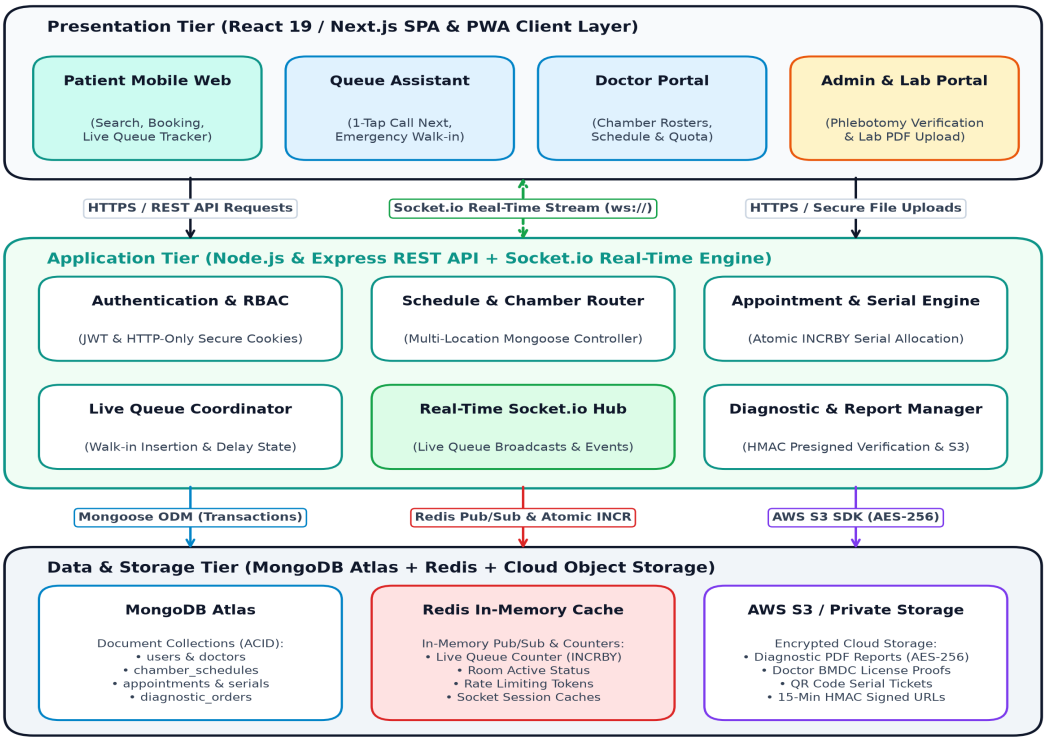


Figure 4.1: SerialSync MERN Stack Multi-Tier Architecture & Real-Time Sync Pipeline

4.1.1 Rationale for Choosing Architectural Model/Style

The selection of the event-driven 3-Tier MERN architecture is grounded in the critical real-time performance requirements of outpatient queuing in high-density urban clinics. Traditional request-response polling (e.g. AJAX polling every 5 seconds) imposes unsustainable server overhead with *O(N)* network traffic scaling per active waiting patient. Socket.io WebSockets establish persistent full-duplex TCP channels, reducing queue state broadcast latency from >5000ms to **<150ms** with minimal bandwidth overhead.

4.2 Technology, Software, and Hardware Used

Architectural Layer	Core Technology	Key Responsibilities & Architectural Role
Presentation Tier	React 19 / Next.js SPA, TailwindCSS, Socket.io Client	High-performance responsive web and mobile PWA clients for Patients, Doctors, Queue Assistants, and Administrators with optimistic UI updates.
Application Tier	Node.js (v20 LTS), Express.js Micro-Routers	Stateless REST API microservices, JWT authentication, RBAC authorization middleware, atomic sequential serial allocation engine, and diagnostic report generator.
Real-Time Event Hub	Socket.io WebSocket Server, Redis Pub/Sub	Bi-directional real-time event broadcasting for live chamber queue state ('Now Serving: #12'), delay notifications, and emergency walk-in broadcasts.
Data & Storage Tier	MongoDB Atlas, Mongoose ODM, AWS S3	Document-oriented ACID multi-document transactional persistence for users, schedules, appointments, and HMAC-SHA256 encrypted digital diagnostic reports.

Table 4.1: MERN Stack Component Responsibilities Breakdown for SerialSync

5. Design

5.1 Component Level Design Following Pattern

SerialSync enforces strict software design patterns across its core subsystems: **Strategy Pattern** for multi-role RBAC authorization, **Factory Pattern** for dynamic chamber slot generation, **Observer & Pub/Sub Pattern** for real-time WebSocket queue synchronization, and **Template Method Pattern** for the multi-stage diagnostic specimen lifecycle.

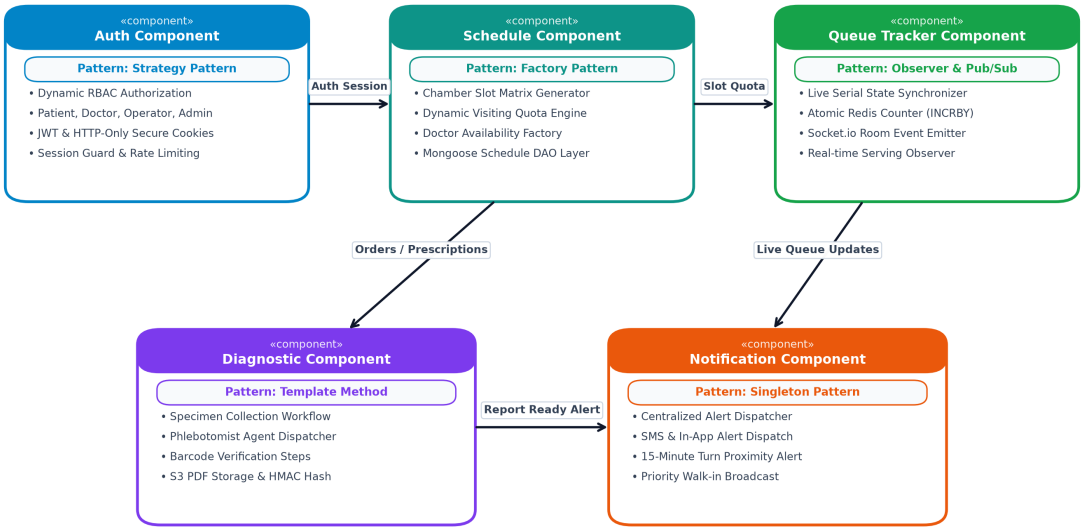


Figure 5.1: Component-Level Design Architecture Following Design Patterns (MERN)

5.2 Dataflow Diagram

Data Flow Diagrams depict how data enters, is processed, and is stored within SerialSync across three levels of decomposition.

5.2.1 DFD Level 0 (System Context Diagram)

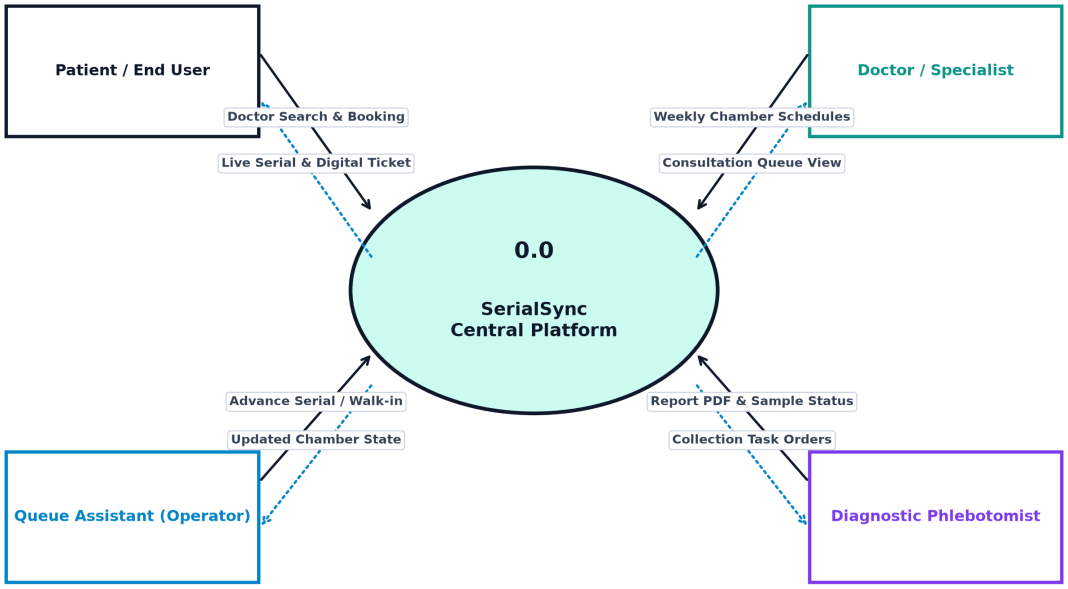


Figure 5.2: Data Flow Diagram (DFD) Level 0 - System Context Diagram

5.2.2 DFD Level 1 (Functional Decomposition)

Figure 5.3 shows the functional decomposition into 5 core processes: 1.0 Auth & Profile, 2.0 Doctor Chamber Scheduling, 3.0 Serial Booking Engine, 4.0 Live Queue Synchronization, and 5.0 Diagnostic Specimen & Report Handling.

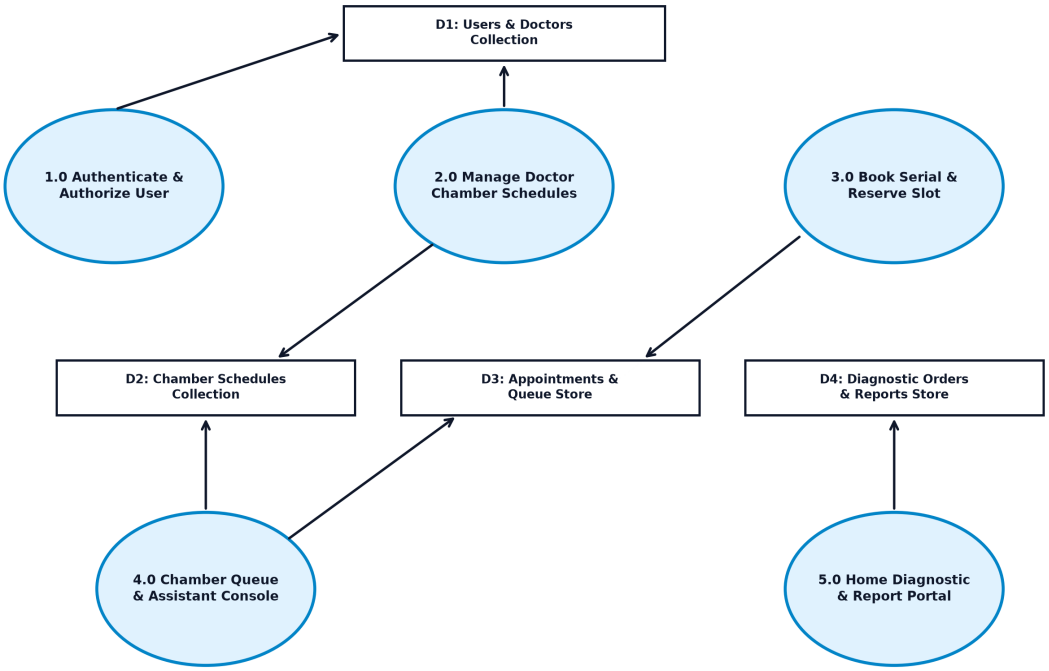


Figure 5.3: Data Flow Diagram (DFD) Level 1 - System Functional Decomposition

5.2.3 DFD Level 2 (Real-Time Queue & Chamber Console Decomposition)

Figure 5.4 details sub-processes for Process 4.0: 4.1 (Validate Assistant Auth), 4.2 (Advance Serving Counter in Redis), 4.3 (Insert Emergency Walk-in Token), and 4.4 (Broadcast Socket.io Event to Patient Room).

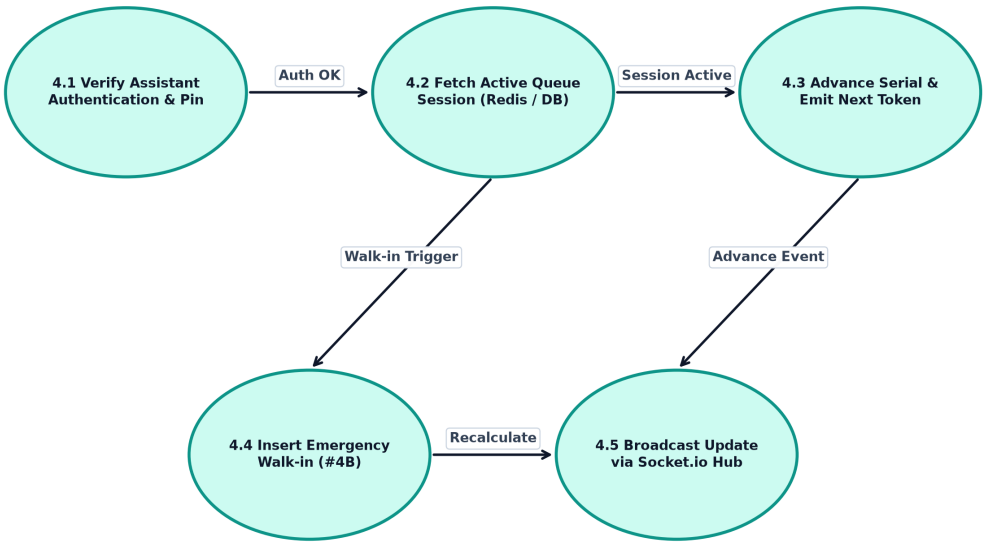


Figure 5.4: Data Flow Diagram (DFD) Level 2 - Live Queue & Walk-in Synchronization

5.2.1 Entity Relationship Diagram

Figure 5.5 presents the comprehensive MongoDB document collection schema and entity relationship model, illustrating relations between `User`, `Doctor`, `ChamberSchedule`, `Appointment`, `QueueSession`, `DiagnosticOrder`, and `AuditLog`.

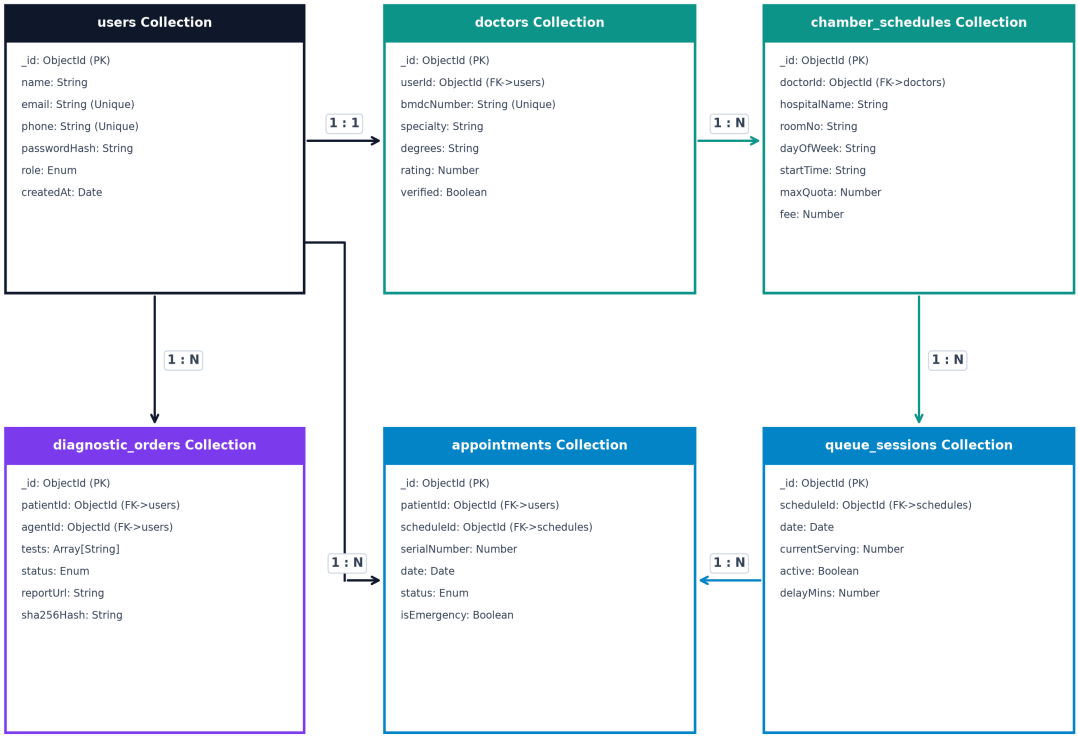


Figure 5.5: MongoDB Document Collection Schema Architecture (MERN)

5.3 GUI (Graphical User Interface) design

The user interface of SerialSync is engineered following a modern, high-contrast dark medical aesthetic (Slate-900 / Sky-600 / Emerald-500) inspired by leading on-demand platforms (Pathao, Uber Health, Linear, Practo). The design incorporates responsive grid layouts, glassmorphism cards, live data badges, real-time WebSocket state synchronizers, and touch-optimized ergonomics for high-velocity clinic operations.

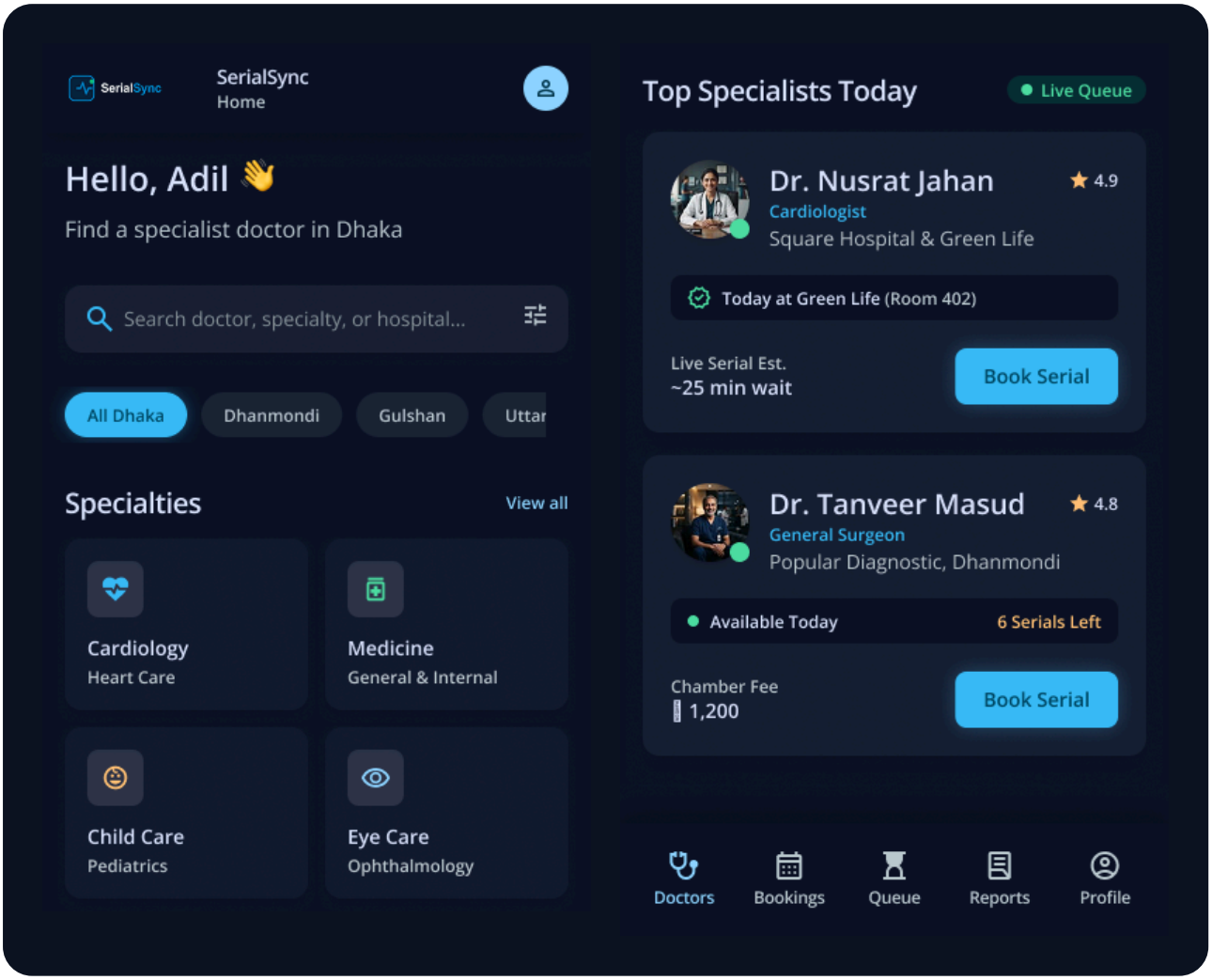
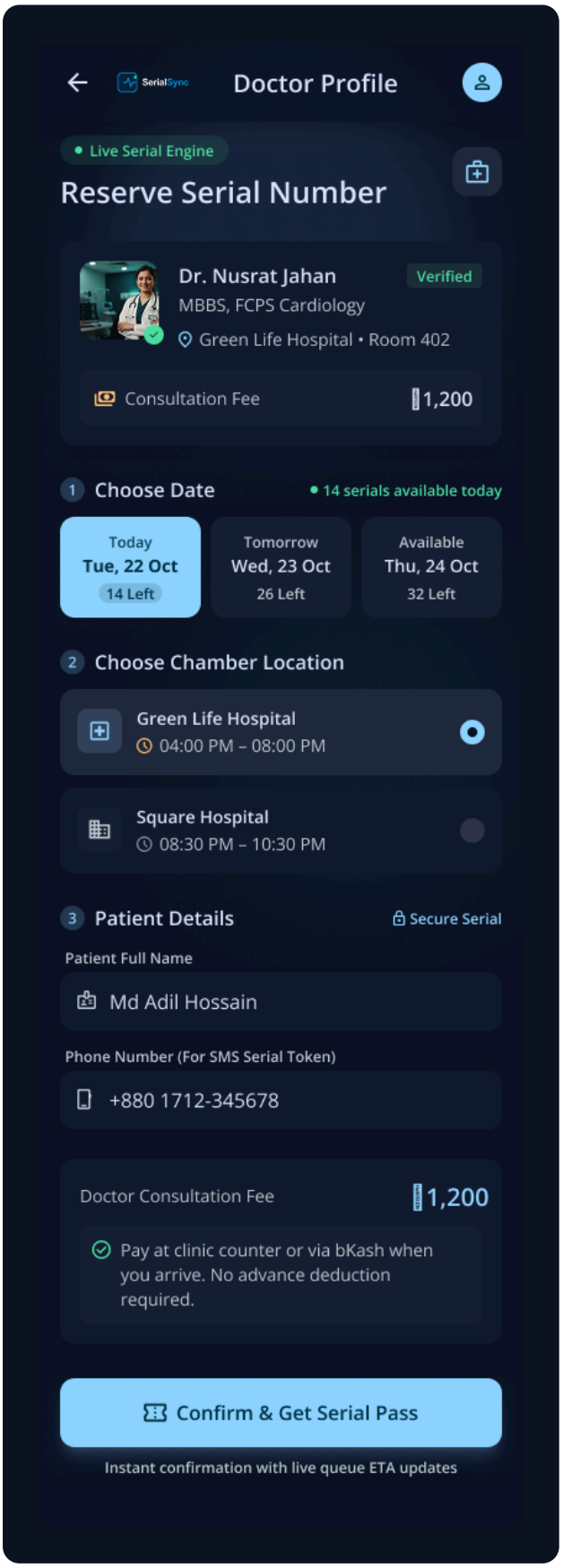


Figure 5.6: GUI Mockup - Patient Specialist Search & Multi-Location Chamber Portal (React 19 / Dark Theme)

Patient Specialist Search & Multi-Location Chamber Portal: The home screen greets the patient and offers a live search bar for doctors, specialties and hospitals, area chips (All Dhaka, Dhanmondi, Gulshan, Uttara) and specialty tiles (Cardiology, Medicine, Child Care, Eye Care). *Top Specialists Today* cards show the doctor photo with an online indicator, rating, hospital and room, live-serial wait estimate, chamber fee and a one-tap *Book Serial* button, with a *Live Queue* badge. A persistent bottom bar (Doctors, Bookings, Queue, Reports, Profile) links the five patient areas.



Doctor Profile & Serial Reservation: The *Reserve Serial Number* screen shows the verified doctor card (MBBS, FCPS Cardiology, hospital, room and consultation fee) and a three-step flow: (1) choose the consultation date with the number of serials left per day, (2) choose the chamber location across hospitals with its time window (e.g. Green Life Hospital 04:00-08:00 PM, Square Hospital 08:30-10:30 PM), and (3) confirm patient name and phone number for the SMS serial token. A fee note explains that payment is made at the clinic counter or via bKash with no advance deduction, and the *Confirm & Get Serial Pass* button issues the serial with live queue ETA updates.

Figure 5.7: GUI Mockup - Doctor Profile, Multi-Chamber Selection & Serial Reservation Portal

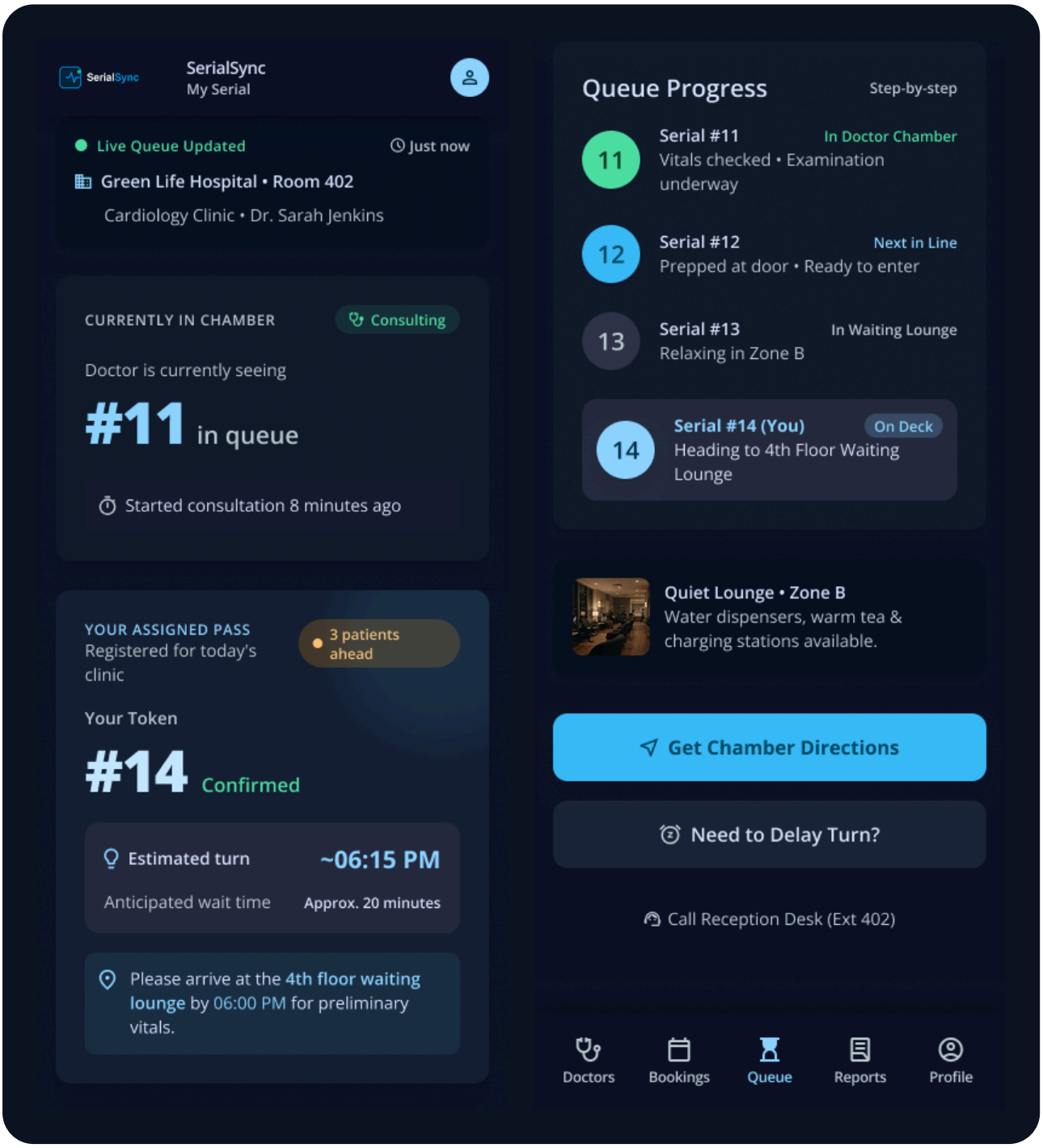
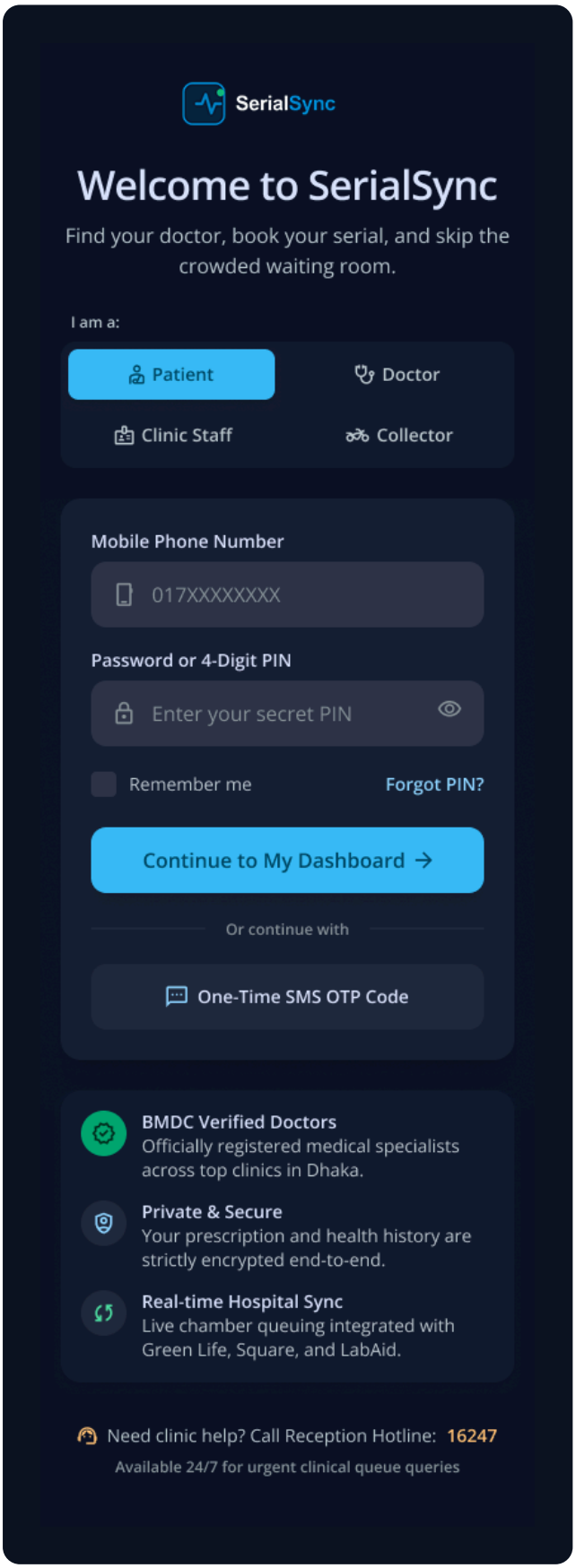


Figure 5.8: GUI Mockup - Patient Real-Time Live Queue Tracker (Socket.io Powered)

Patient Real-Time Live Queue Tracker: Mobile tracking interface fed by Socket.io WebSocket streaming (shown in two halves, top then bottom). It displays a *Live Queue Updated* status with hospital and room, the *Currently in Chamber* serial (#11) and the patient's assigned pass (#14, Confirmed) with patients ahead, estimated turn time (about 06:15 PM, approx. 20 minutes) and arrival instructions. A step-by-step *Queue Progress* list marks serials #11 (in doctor chamber), #12 (next in line), #13 (waiting lounge) and #14 (you, on deck). Actions: *Get Chamber Directions*, *Need to Delay Turn?* and a call to the reception desk.



Authentication & Role Selection Portal: The *Welcome to SerialSync* screen lets every user choose a role (Patient, Doctor, Clinic Staff or Collector) before signing in, so the correct RBAC workspace opens. Credentials are a mobile phone number with a password or 4-digit PIN, a *Remember me* option, *Forgot PIN?* recovery and a one-time SMS OTP alternative, followed by the *Continue to My Dashboard* action. Trust panels highlight BMDC-verified doctors, end-to-end private and secure health data, and real-time hospital queuing, and a 24/7 reception hotline is shown for urgent queue queries. Clinic staff use the same entry point to reach the queue assistant console.

Figure 5.9: GUI Mockup - Authentication, Role Selection & Secure Sign-in Portal

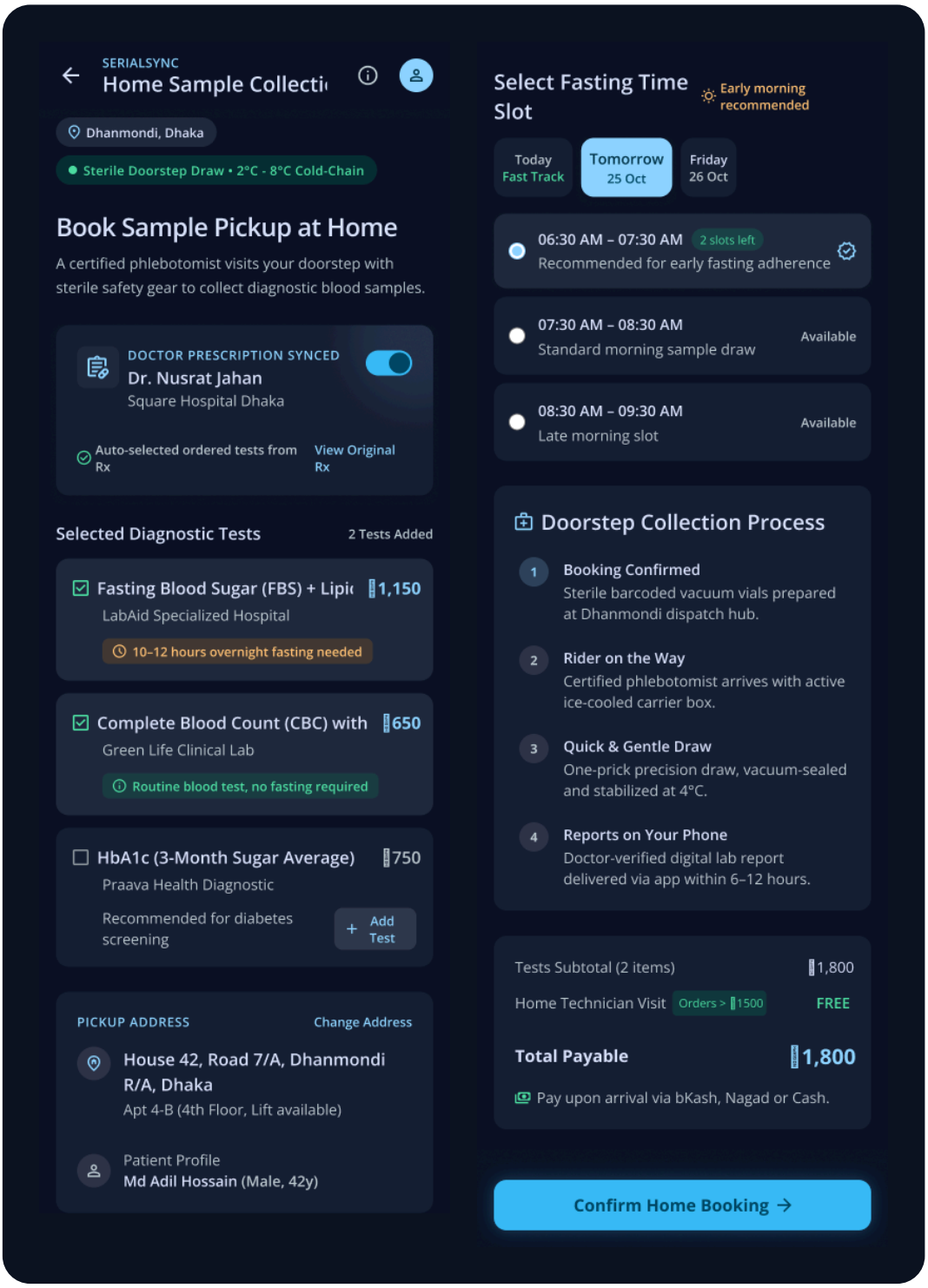


Figure 5.10: GUI Mockup - Home Diagnostic Sample Collection Booking Portal

Home Sample Collection Booking: The patient books a sterile doorstep blood-sample pickup (shown in two halves, top then bottom). A synced doctor prescription auto-selects the ordered tests, and extra tests such as HbA1c can be added; each test lists its lab, preparation note (e.g. 10-12 hour fasting) and price. The patient confirms the pickup address and patient profile, picks a fasting-friendly time slot (early morning recommended) and reviews the four-step doorstep process: booking confirmed, rider on the way, quick draw, and reports on the phone within 6-12 hours. A price summary (subtotal, free home visit above the threshold, total payable) and the *Confirm Home Booking* button complete the flow; payment is by bKash, Nagad or cash on arrival.



Figure 5.11: GUI Mockup - Pathao-Style Phlebotomist Rider & Sample Dispatch Portal

Pathao-Style Phlebotomist Rider & Sample Dispatch Portal: Logistics interface for the certified phlebotomist rider (shown in two halves, top then bottom). The header carries the rider profile, rating and cluster with cold-chain telemetry (cooler temperature, battery, digital lock seal). The active job card shows the priority fasting-draw patient, time slot, address with floor and lift notes, one-tap *Call Patient* and *WhatsApp* actions, and a live route map with ETA and traffic notes. *Dispatch Progress* tracks assigned, on the way, blood draw and barcode scan, and delivery to the central lab, while the *Required Kit Checklist* lists colour-coded tubes with barcode IDs, a camera barcode scan, *I Have Arrived* and *Report Traffic Delay* actions.

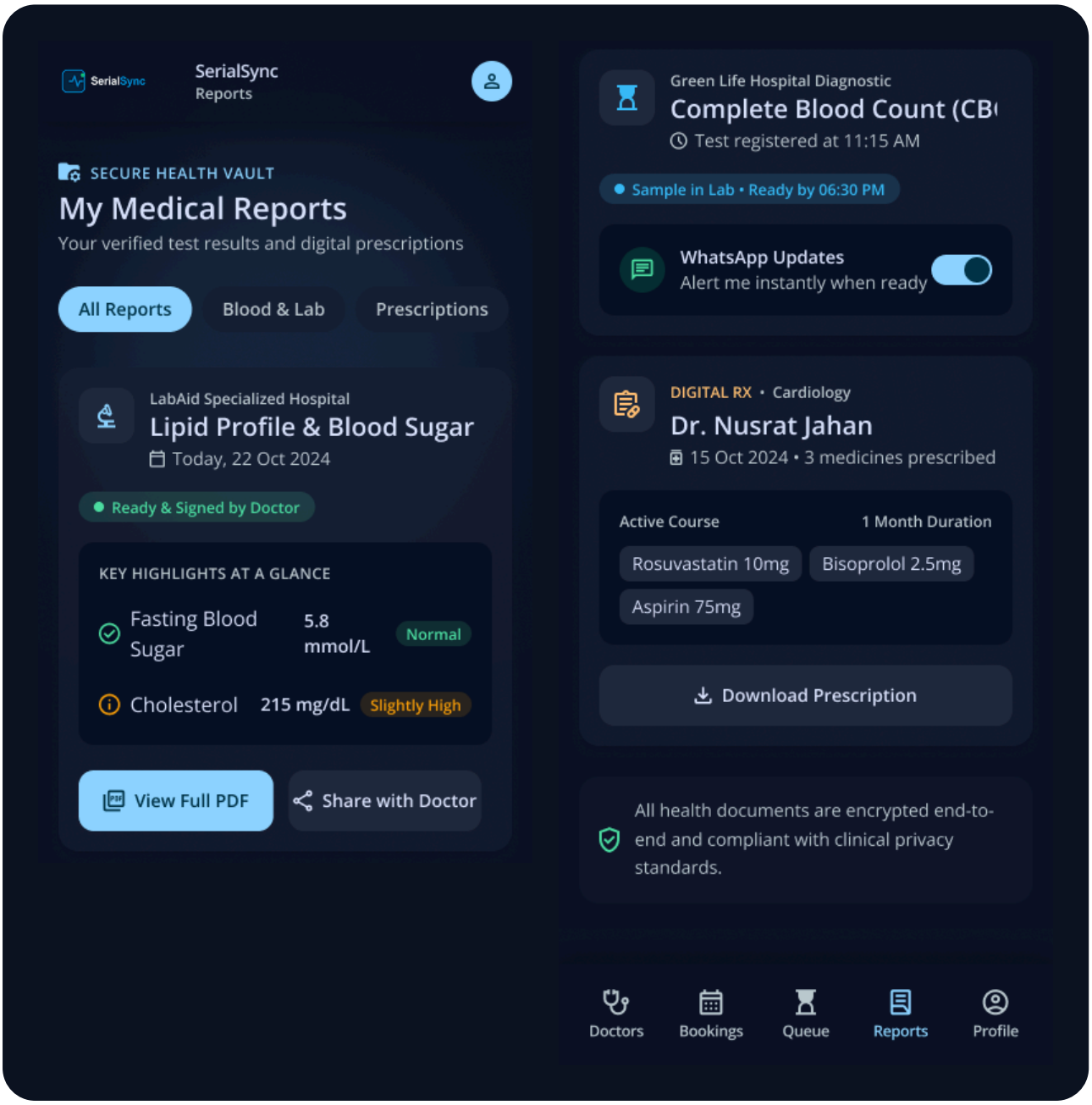


Figure 5.12: GUI Mockup - Secure Medical Reports Vault & Digital Prescriptions Portal

Secure Medical Reports Vault & Digital Prescriptions: The *My Medical Reports* screen (shown in two halves, top then bottom) is the patient's secure health vault with filters for All Reports, Blood & Lab and Prescriptions. Each signed report card shows the lab, test name, date and status (*Ready & Signed by Doctor* or *Sample in Lab*), key highlights with Normal / Slightly High flags, *View Full PDF* and *Share with Doctor* actions, and a WhatsApp alert toggle for pending results. Digital prescriptions list the active medicines and course duration with a *Download Prescription* action, and a footer reminds users that all health documents are encrypted end-to-end under clinical privacy standards.

6. API and Testing Documentation

SerialSync exposes a RESTful API together with a bi-directional WebSocket interface (Socket.io) over HTTP(S) and WSS. All REST endpoints consume and produce JSON (application/json) except multipart report upload and PDF download. Standard HTTP status codes are used: 200 OK, 201 Created, 400 Bad Request, 401 Unauthorized, 403 Forbidden, 404 Not Found, 409 Conflict, and 500 Internal Server Error.

This chapter documents every endpoint (Sections 6.1 to 6.7) and the automated pass and fail test cases that verify them (Sections 6.8 to 6.11). Paths below are the implemented Express 5 routes under /api/* (not /api/v1/*). Successful responses use { success: true, ... }. Errors use { success: false, message: "..." }. Session is an HTTP-only cookie named serialsync_token (7-day JWT) or Authorization: Bearer <token>.

Conventions

HTTP code	Meaning
200 OK	Successful GET / PATCH / POST mutation without a new resource
201 Created	Resource created (register, chamber, schedule, booking, walk-in, order)
400 Bad Request	Validation failure (missing fields, invalid enum, weak password)
401 Unauthorized	Missing or invalid session cookie / Bearer token
403 Forbidden	Role or doctor-approval gate (DOCTOR_PENDING)
404 Not Found	Unknown id
409 Conflict	Duplicate booking or overlapping schedule

Role	Typical access
patient	Book serials, live queue, diagnostic orders, profile
doctor	Chambers, schedules, day itinerary (after admin approval)
assistant	Queue console: advance, walk-in, delay, pause
phlebotomist	Assigned diagnostic dispatch queue and status
admin	Doctor BMDC approval, reports upload, all privileged routes

6.1 Module 1: Authentication & User Management

GET /api/health

Access: Public

Description: Service liveness probe used by deployment health checks.

Parameter	Type	Required	Description
?	?	No	No parameters.

Sample Request Body

```
None (GET Request)
```

Sample JSON Response

```
{
  "success": true,
  "service": "serialsync-api",
  "status": "ok"
}
```

POST /api/auth/register

Access: Public (All Users)

Description: Registers a new account. Password is hashed with bcrypt (12 salt rounds). Doctors are created with doctorApprovalStatus = pending. Sets the serialsync_token cookie.

Parameter	Type	Required	Description
fullName	String	Yes	Full legal name of the user.
email	String	Yes	Unique email used for login and notifications.
password	String	Yes	Minimum 8 characters.
phone	String	No	Bangladeshi mobile number (01XXXXXXXX).
role	Enum	No	patient (default), doctor, assistant, phlebotomist, admin (admin blocked in production).
doctorProfile	Object	No	Optional specialtyTitle, specialtySlug, affiliations (doctors).

Sample Request Body

```
{
  "fullName": "Md Adil Hossain",
  "email": "adil@example.com",
  "phone": "01712345678",
  "password": "SecurePassword123",
  "role": "patient"
}
```

Sample JSON Response (201)

```
{
  "success": true,
  "user": {
    "id": "68f1a2b3c4d5e6f708091011",
    "fullName": "Md Adil Hossain",
    "email": "adil@example.com",
    "phone": "01712345678",
    "role": "patient"
  }
}
```

POST /api/auth/login

Access: Public (All Users)

Description: Authenticates with email and password; returns the user and sets the HTTP-only JWT cookie.

Parameter	Type	Required	Description
email	String	Yes	Registered account email.
password	String	Yes	Account plaintext password.

Sample Request Body

```
{
  "email": "adil@example.com",
  "password": "SecurePassword123"
}
```

Sample JSON Response (200)

```
{
  "success": true,
  "user": {
    "id": "68f1a2b3c4d5e6f708091011",
    "fullName": "Md Adil Hossain",
    "email": "adil@example.com",
    "role": "patient"
  }
}
```

POST /api/auth/logout

Access: Authenticated User

Description: Clears the session cookie. Subsequent protected requests return 401.

Parameter	Type	Required	Description
Cookie	Header	Yes	serialsync_token session cookie.

Sample Request Body

```
None (POST with Cookie)
```

Sample JSON Response

```
{
  "success": true,
  "message": "Logged out"
}
```

GET /api/auth/me

Access: Authenticated User

Description:

Returns the currently authenticated user from the JWT session.

Parameter	Type	Required	Description
Authorization	Header	Yes	Cookie serialsync_token or Bearer token.

Sample Request Body

```
None (GET Request with Cookie/Header)
```

Sample JSON Response

```
{
  "success": true,
  "user": {
    "id": "68f1a2b3c4d5e6f708091011",
    "fullName": "Md Adil Hossain",
    "role": "patient"
  }
}
```

POST /api/auth/forgot-password

Access: Public

Description: Starts password reset. In non-production, the response includes resetToken for QA.

Parameter	Type	Required	Description
email	String	Yes	Registered email address.

Sample Request Body

```
{
  "email": "adil@example.com"
}
```

Sample JSON Response

```
{
  "success": true,
  "message": "If that email exists, a reset link was sent",
  "resetToken": "a1b2c3d4..."
}
```

POST /api/auth/reset-password

Access: Public

Description: Completes password reset using the token from forgot-password.

Parameter	Type	Required	Description
token	String	Yes	One-time reset token.
password	String	Yes	New password (min 8 characters).

Sample Request Body

```
{
  "token": "a1b2c3d4...",
  "password": "newpassword99"
}
```

Sample JSON Response

```
{
  "success": true,
  "message": "Password updated"
}
```

POST /api/auth/change-password

Access: Authenticated User

Description: Changes password for the logged-in user after verifying the current password.

Parameter	Type	Required	Description
currentPassword	String	Yes	Existing password.
newPassword	String	Yes	New password (min 8 characters).

Sample Request Body

```
{
  "currentPassword": "SecurePassword123",
  "newPassword": "EvenSaferPass99"
}
```

Sample JSON Response

```
{
  "success": true,
  "message": "Password changed"
}
```

GET /api/users/me

Access: Authenticated User

Description: Retrieves the logged-in profile including patientProfile or doctorProfile.

Parameter	Type	Required	Description
Authorization	Header	Yes	Session cookie or Bearer token.

Sample Request Body

```
None (GET Request with Cookie/Header)
```

Sample JSON Response

```
{
  "success": true,
  "user": {
    "id": "68f1a2b3c4d5e6f708091011",
    "fullName": "Md Adil Hossain",
    "email": "adil@example.com",
    "phone": "01712345678",
    "role": "patient",
    "patientProfile": {
      "city": "Dhaka",
      "bloodType": "B+",
      "smartPassVerified": true
    }
  }
}
```

PATCH /api/users/me

Access: Authenticated User

Description: Updates fullName, phone, and role-specific profile fields.

Parameter	Type	Required	Description
fullName	String	No	Updated display name.
phone	String	No	Updated mobile number.
patientProfile	Object	No	city, bloodType, smartPassVerified (patients).
doctorProfile	Object	No	specialtyTitle, affiliations, roomLabel (doctors).

Sample Request Body

```
{
  "fullName": "Updated Name",
  "phone": "01811111111",
  "patientProfile": { "city": "Dhaka" }
}
```

Sample JSON Response

```
{
  "success": true,
  "user": {
    "id": "68f1a2b3c4d5e6f708091011",
    "fullName": "Updated Name",
    "phone": "01811111111",
    "role": "patient"
  }
}
```

GET /api/users/admin/overview

Access: System Administrator

Description: RBAC smoke endpoint confirming admin session.

Parameter	Type	Required	Description
Authorization	Header	Yes	Admin session.

Sample Request Body

```
None (GET Request)
```

Sample JSON Response

```
{
  "success": true,
  "message": "Admin RBAC route",
  "admin": { "id": "...", "role": "admin" }
}
```

6.2 Module 2: Doctor Multi-Location Schedules

GET **/api/doctors**

Access: Authenticated User

Description: Searches the public directory of approved doctors by specialty, city, and free-text query.

Parameter	Type	Required	Description
q	Query	No	Name / affiliation search string.
specialty	Query	No	Specialty slug (e.g. cardiology).
city	Query	No	City filter (e.g. Dhaka).

Sample Request Body

```
None (Query Parameters)
```

Sample JSON Response

```
{
  "success": true,
  "doctors": [{
    "id": "68f1doc...",
    "fullName": "Prof. Dr. M. Rahman",
    "doctorProfile": {
      "specialtyTitle": "Cardiologist",
      "affiliations": "Green Life Hospital"
    }
  }],
  "meta": { "specialties": ["cardiology", "medicine"] }
}
```

GET **/api/doctors/:doctorId**

Access: Authenticated User

Description: Returns a doctor profile with chambers and active weekly schedules for the booking UI.

Parameter	Type	Required	Description
doctorId	Path	Yes	Doctor User ObjectId.

Sample Request Body

```
None (GET Request)
```

Sample JSON Response

```
{
  "success": true,
  "doctor": {
    "id": "68f1doc...",
    "fullName": "Prof. Dr. M. Rahman",
    "doctorApprovalStatus": "approved"
  },
  "chambers": [{ "id": "ch_01", "name": "Green Life", "area": "Dhanmon
  "schedules": [{ "id": "sch_01", "daysOfWeek": [1, 3], "startTime": "
}
```

POST **/api/chambers**

Access: Verified Doctor (approved)

Description: Creates a hospital or private clinic chamber owned by the authenticated doctor. Pending doctors receive 403 DOCTOR_PENDING.

Parameter	Type	Required	Description
name	String	Yes	Chamber / clinic display name.
address	String	No	Street address.
city	String	No	City (default Dhaka).
area	String	No	Area (e.g. Dhanmondi).
phone	String	No	Chamber contact number.

Sample Request Body

```
{
  "name": "Green Life Hospital",
  "address": "32 Green Road",
  "city": "Dhaka",
  "area": "Dhanmondi"
}
```

Sample JSON Response (201)

```
{
  "success": true,
  "chamber": {
    "id": "68f1ch...",
    "name": "Green Life Hospital",
    "area": "Dhanmondi",
    "isActive": true
  }
}
```

GET /api/chambers

Access: Verified Doctor (approved)

Description: Lists chambers owned by the authenticated doctor.

Parameter	Type	Required	Description
Authorization	Header	Yes	Doctor session.

Sample Request Body

```
None (GET Request)
```

Sample JSON Response

```
{
  "success": true,
  "chambers": [{ "id": "68f1ch...", "name": "Green Life Hospital" }]
}
```

PATCH /api/chambers/:id

Access: Verified Doctor (approved)

Description: Updates chamber metadata (name, address, area, phone, isActive).

Parameter	Type	Required	Description
id	Path	Yes	Chamber ObjectId.
name	String	No	Updated facility name.
isActive	Boolean	No	Soft-disable chamber.

Sample Request Body

```
{
  "phone": "01711112222",
  "isActive": true
}
```

Sample JSON Response

```
{
  "success": true,
  "chamber": { "id": "68f1ch...", "phone": "01711112222" }
}
```

GET /api/chambers/directory

Access: Queue Assistant, Administrator

Description: Cross-doctor chamber directory for assistant console session setup.

Parameter	Type	Required	Description
Authorization	Header	Yes	Assistant or admin session.

Sample Request Body

```
None (GET Request)
```

Sample JSON Response

```
{
  "success": true,
  "chambers": [{
    "id": "68f1ch...",
    "name": "Green Life Hospital",
    "doctorId": "68f1doc..."
  }]
}
```

POST /api/schedules

Access: Verified Doctor (approved)

Description:

Creates a recurring weekly availability at a chamber with overlap detection (409 on conflict).

Parameter	Type	Required	Description
chamberId	String	Yes	Owned Chamber ObjectId.
daysOfWeek	Integer[]	Yes	0=Sun ? 6=Sat.
startTime	String	Yes	24h start (HH:MM).
endTime	String	Yes	24h end (HH:MM), after startTime.
consultationFee	Number	Yes	Fee in BDT.
slotDurationMinutes	Integer	Yes	Minutes per serial (5?120).
patientsPerSlot	Integer	No	Quota per slot (default 1).
notes	String	No	Optional chamber notes.

Sample Request Body

```
{
  "chamberId": "68f1ch...",
  "daysOfWeek": [1, 3],
  "startTime": "18:00",
  "endTime": "21:00",
  "consultationFee": 1200,
  "patientsPerSlot": 2,
  "slotDurationMinutes": 15
}
```

Sample JSON Response (201)

```
{
  "success": true,
  "schedule": {
    "id": "68f1sch...",
    "doctorId": "68f1doc...",
    "daysOfWeek": [1, 3],
    "startTime": "18:00",
    "endTime": "21:00",
    "status": "active"
  }
}
```

GET /api/schedules/me

Access: Verified Doctor (approved)

Description: Lists the doctor's own schedules including inactive rows.

Parameter	Type	Required	Description
Authorization	Header	Yes	Doctor session.

Sample Request Body

```
None (GET Request)
```

Sample JSON Response

```
{
  "success": true,
  "schedules": [{ "id": "68f1sch...", "status": "active" }]
}
```

GET /api/schedules/doctors/:doctorId

Access: Public

Description: Public active schedules and chambers for a doctor. Inactive schedules are omitted.

Parameter	Type	Required	Description
doctorId	Path	Yes	Doctor User ObjectId.

Sample Request Body

```
None (GET Request)
```

Sample JSON Response

```
{
  "success": true,
  "schedules": [{ "id": "68f1sch...", "daysOfWeek": [1, 3] }],
  "chambers": [{ "id": "68f1ch...", "name": "Green Life Hospital" }]
}
```

GET /api/schedules/doctors/:doctorId/slots

Access: Public

Description: Expands weekly schedules into bookable slots for a date range, with remaining quota per slot.

Parameter	Type	Required	Description
doctorId	Path	Yes	Doctor User ObjectId.
from	Query	Yes	Start date YYYY-MM-DD.
to	Query	Yes	End date YYYY-MM-DD.

Sample Request Body

```
None (Query Parameters)
```

Sample JSON Response

```
{
  "success": true,
  "slots": [{
    "scheduleId": "68f1sch...",
    "date": "2026-10-05",
    "slotStart": "18:00",
    "slotEnd": "18:15",
    "remaining": 2
  }]
}
```

PATCH /api/schedules/:id

Access: Verified Doctor (approved)

Description: Updates times, days, fee, or chamber. Overlaps return 409.

Parameter	Type	Required	Description
id	Path	Yes	Schedule ObjectId.
startTime	String	No	Updated 24h start.
endTime	String	No	Updated 24h end.
daysOfWeek	Integer[]	No	Updated weekday set.
consultationFee	Number	No	Updated fee in BDT.

Sample Request Body

```
{
  "startTime": "18:30",
  "endTime": "21:30",
  "consultationFee": 1500
}
```

Sample JSON Response

```
{
  "success": true,
  "schedule": {
    "id": "68f1sch...",
    "startTime": "18:30",
    "consultationFee": 1500
  }
}
```

PATCH /api/schedules/:id/status

Access: Verified Doctor (approved)

Description: Toggles schedule visibility. Inactive schedules disappear from the public booking view.

Parameter	Type	Required	Description
id	Path	Yes	Schedule ObjectId.
status	Enum	Yes	active or inactive.

Sample Request Body

```
{
  "status": "inactive"
}
```

Sample JSON Response

```
{
  "success": true,
  "schedule": { "id": "68f1sch...", "status": "inactive" }
}
```

DELETE /api/schedules/:id

Access: Verified Doctor (approved)

Description: Deletes an availability row owned by the doctor.

Parameter	Type	Required	Description
id	Path	Yes	Schedule ObjectId.

Sample Request Body

```
None (DELETE Request)
```

Sample JSON Response

```
{
  "success": true,
  "message": "Schedule deleted"
}
```

6.3 Module 3: Appointment Booking & Dynamic Serial Allocation

POST **/api/schedules/slots/book**

Access: Authenticated Patient, Administrator

Description: Atomically reserves a sequential serial for the doctor-day (Redis INCR with in-memory fallback) and creates a SlotBooking with confirmationCode.

Parameter	Type	Required	Description
scheduleId	String	Yes	Target DoctorSchedule ObjectId.
date	String	Yes	Appointment date YYYY-MM-DD.
slotStart	String	Yes	Slot start time HH:MM from the slots API.
patientNote	String	No	Optional note for the doctor.

Sample Request Body

```
{
  "scheduleId": "68f1sch...",
  "date": "2026-10-05",
  "slotStart": "18:00"
}
```

Sample JSON Response (201)

```
{
  "success": true,
  "appointment": {
    "id": "68f1apt...",
    "serialNumber": 14,
    "confirmationCode": "SS-A1B2C3",
    "status": "booked",
    "date": "2026-10-05",
    "slotStart": "18:00"
  }
}
```

GET **/api/appointments/me**

Access: Authenticated Patient, Administrator

Description: Chronological list of the patient's bookings (optional status filter).

Parameter	Type	Required	Description
status	Query	No	booked, confirmed, completed, cancelled, no_show.

Sample Request Body

```
None (GET Request)
```

Sample JSON Response

```
{
  "success": true,
  "appointments": [{
    "id": "68f1apt...",
    "serialNumber": 14,
    "status": "booked",
    "doctor": { "fullName": "Prof. Dr. M. Rahman" },
    "chamber": { "name": "Green Life Hospital" }
  }]
}
```

GET **/api/appointments/doctor/day**

Access: Verified Doctor, Administrator

Description: Doctor chamber itinerary for a calendar day.

Parameter	Type	Required	Description
date	Query	No	YYYY-MM-DD (defaults to today).

Sample Request Body

```
None (Query Parameters)
```

Sample JSON Response

```
{
  "success": true,
  "date": "2026-10-05",
  "appointments": [{
    "serialNumber": 1,
    "slotStart": "18:00",
    "status": "booked"
  }]
}
```

GET /api/appointments/:id

Access: Owner Patient, Assigned Doctor, Administrator

Description: Appointment detail. Other patients receive 403.

Parameter	Type	Required	Description
id	Path	Yes	SlotBooking ObjectId.

Sample Request Body

```
None (GET Request)
```

Sample JSON Response

```
{
  "success": true,
  "appointment": {
    "id": "68f1apt...",
    "serialNumber": 14,
    "confirmationCode": "SS-A1B2C3",
    "status": "booked"
  }
}
```

GET /api/appointments/:id/qr

Access: Owner Patient, Administrator

Description: Returns a PNG data URL QR encoding confirmationCode, serialNumber, and chamber metadata (SmartPass).

Parameter	Type	Required	Description
id	Path	Yes	SlotBooking ObjectId.

Sample Request Body

```
None (GET Request)
```

Sample JSON Response

```
{
  "success": true,
  "qrDataUrl": "...",
  "payload": {
    "appointmentId": "68f1apt...",
    "confirmationCode": "SS-A1B2C3",
    "serialNumber": 14,
    "date": "2026-10-05"
  }
}
```

POST /api/appointments/:id/cancel

Access: Owner Patient, Administrator

Description: Cancels a booked/confirmed appointment. Completed visits cannot be cancelled (400).

Parameter	Type	Required	Description
id	Path	Yes	SlotBooking ObjectId.

Sample Request Body

```
None (POST Request)
```

Sample JSON Response

```
{
  "success": true,
  "appointment": {
    "id": "68f1apt...",
    "status": "cancelled"
  }
}
```

PATCH /api/appointments/:id/status

Access: Assigned Doctor, Administrator

Description: Doctor updates visit workflow status with allowed transitions.

Parameter	Type	Required	Description
id	Path	Yes	SlotBooking ObjectId.
status	Enum	Yes	confirmed, completed, no_show, cancelled.

Sample Request Body

```
{
  "status": "completed"
}
```

Sample JSON Response

```
{
  "success": true,
  "appointment": { "id": "68f1apt...", "status": "completed" }
}
```

6.4 Module 4: Queue Assistant & Chamber Operations

Queue mutations emit Socket.io event queue_updated to room queue:<sessionId>. Clients connect to path /socket.io with the same JWT cookie (or handshake auth.token), then emit join_queue { sessionId }.

POST /api/queue/sessions

Access: Queue Assistant, Doctor, Administrator

Description: Ensures a QueueSession for doctor + chamber + date and returns the built queue state.

Parameter	Type	Required	Description
doctorId	String	Yes	Doctor User ObjectId.
chamberId	String	Yes	Chamber ObjectId.
date	String	Yes	Session date YYYY-MM-DD.

Sample Request Body

```
{
  "doctorId": "68f1doc...",
  "chamberId": "68f1ch...",
  "date": "2026-10-05"
}
```

Sample JSON Response

```
{
  "success": true,
  "queue": {
    "sessionId": "68f1qs...",
    "date": "2026-10-05",
    "status": "open",
    "servingIndex": -1,
    "queueLength": 12,
    "nowServing": null,
    "items": [{ "label": "1", "kind": "booking" }]
  }
}
```

GET /api/queue/sessions/:id

Access: Authenticated User

Description: Full queue snapshot: nowServing, upNext, delay, pause, and item list.

Parameter	Type	Required	Description
id	Path	Yes	QueueSession ObjectId.

Sample Request Body

```
None (GET Request)
```

Sample JSON Response

```
{
  "success": true,
  "queue": {
    "sessionId": "68f1qs...",
    "nowServing": { "label": "11", "kind": "booking" },
    "upNext": [{ "label": "12" }],
    "delayMinutes": 0
  }
}
```

GET /api/queue/me/today

Access: Authenticated Patient

Description: Live position for the patient's appointment on the current local date, including patientsAhead and ETA.

Parameter	Type	Required	Description
Authorization	Header	Yes	Patient session.

Sample Request Body

```
None (GET Request)
```

Sample JSON Response

```
{
  "success": true,
  "appointmentId": "68f1apt...",
  "serialNumber": 14,
  "patientsAhead": 3,
  "estimatedWaitMinutes": 45,
  "queue": { "sessionId": "68f1qs...", "nowServing": { "label": "11" }
}
```

POST /api/queue/sessions/:id/advance

Access: Queue Assistant, Administrator

Description: Advances servingIndex by +1 and emits queue_updated to waiting patients.

Parameter	Type	Required	Description
id	Path	Yes	QueueSession ObjectId.

Sample Request Body

```
None (POST Request)
```

Sample JSON Response

```
{
  "success": true,
  "queue": {
    "sessionId": "68f1qs...",
    "servingIndex": 0,
    "nowServing": { "label": "1", "kind": "booking" }
  }
}
```

POST /api/queue/sessions/:id/walk-in

Access: Queue Assistant, Administrator

Description: Injects a walk-in with sub-index (e.g. 11B) after afterSerial without renumbering later bookings.

Parameter	Type	Required	Description
id	Path	Yes	QueueSession ObjectId.
afterSerial	Integer	Yes	Insert after this booked serial.
patientName	String	Yes	Walk-in display name.
priority	Enum	No	emergency | urgent | standard.
reason	String	No	Optional clinical reason.

Sample Request Body

```
{
  "afterSerial": 11,
  "patientName": "Rahim Uddin",
  "priority": "emergency"
}
```

Sample JSON Response (201)

```
{
  "success": true,
  "walkIn": {
    "displayLabel": "11B",
    "patientName": "Rahim Uddin"
  },
  "queue": { "queueLength": 13 }
}
```

POST /api/queue/sessions/:id/no-show

Access: Queue Assistant, Administrator

Description: Marks a booked patient as no-show and skips them in the serving order.

Parameter	Type	Required	Description
id	Path	Yes	QueueSession ObjectId.
bookingId	String	Yes	SlotBooking ObjectId.

Sample Request Body

```
{
  "bookingId": "68f1apt..."
}
```

Sample JSON Response

```
{
  "success": true,
  "queue": { "sessionId": "68f1qs..." }
}
```

POST /api/queue/sessions/:id/undo

Access: Queue Assistant, Administrator

Description: Reverses the last no-show action on the session.

Parameter	Type	Required	Description
id	Path	Yes	QueueSession ObjectId.

Sample Request Body

```
None (POST Request)
```

Sample JSON Response

```
{
  "success": true,
  "queue": { "sessionId": "68f1qs..." }
}
```

POST /api/queue/sessions/:id/delay

Access: Queue Assistant, Administrator

Description: Sets delayMinutes / delayReason; clients recompute ETA from queue_updated.

Parameter	Type	Required	Description
id	Path	Yes	QueueSession ObjectId.
delayMinutes	Integer	Yes	Estimated extra wait (e.g. 15, 30, 45).
delayReason	String	No	e.g. Traffic on Mirpur Road.

Sample Request Body

```
{
  "delayMinutes": 30,
  "delayReason": "Traffic on Mirpur Road"
}
```

Sample JSON Response

```
{
  "success": true,
  "queue": {
    "delayMinutes": 30,
    "delayReason": "Traffic on Mirpur Road"
  }
}
```

POST /api/queue/sessions/:id/pause

Access: Queue Assistant, Administrator

Description:

Pauses or resumes the chamber session with an optional public message.

Parameter	Type	Required	Description
id	Path	Yes	QueueSession ObjectId.
paused	Boolean	Yes	true to pause, false to resume.
pauseMessage	String	No	Message shown on the live tracker.

Sample Request Body

```
{
  "paused": true,
  "pauseMessage": "Doctor in emergency OT"
}
```

Sample JSON Response

```
{
  "success": true,
  "queue": {
    "status": "paused",
    "pauseMessage": "Doctor in emergency OT"
  }
}
```

6.4.1 WebSocket contract (Socket.io)

Event	Direction	Payload	Description
join_queue	Client ? Server	{ sessionId }	Subscribe to room queue:<sessionId>.
leave_queue	Client ? Server	{ sessionId }	Unsubscribe from the room.
queue_updated	Server ? Client	Queue state object	Emitted after REST queue mutations.

Handshake authentication: cookie serialsync_token or socket.handshake.auth.token. Path: /socket.io. CORS origin: CLIENT_URL.

6.5 Module 5: Home Diagnostics & Reports

GET **/api/diagnostics/tests**

Access: Authenticated User

Description: Active diagnostic test catalog (CBC, lipid, HbA1c, etc.).

Parameter	Type	Required	Description
search	Query	No	Name or code substring.
category	Query	No	Category filter or all.

Sample Request Body

```
None (Query Parameters)
```

Sample JSON Response

```
{
  "success": true,
  "tests": [{
    "id": "68f1t...",
    "code": "CBC",
    "name": "Complete Blood Count",
    "category": "Hematology",
    "price": 450
  }]
}
```

GET **/api/diagnostics/packages**

Access: Authenticated User

Description: Bundled executive / diabetic packages with discount metadata.

Parameter	Type	Required	Description
Authorization	Header	Yes	Session cookie.

Sample Request Body

```
None (GET Request)
```

Sample JSON Response

```
{
  "success": true,
  "packages": [{
    "id": "pkg-executive",
    "name": "Executive Health Check",
    "discountPercent": 15,
    "testCodes": ["CBC", "LIPID", "LFT", "FBS", "HBA1C"]
  }]
}
```

POST /api/diagnostics/orders

Access: Authenticated Patient

Description: Creates a home collection order. If a phlebotomist exists, status becomes assigned with etaMinutes.

Parameter	Type	Required	Description
testIds	Array	Conditional	DiagnosticTest ObjectIds (required if no packageId).
packageId	String	Conditional	pkg-executive or pkg-diabetic.
address	String	Yes	Residential collection address.
contactPhone	String	Yes	Contact mobile number.
preferredDate	String	Yes	Collection date YYYY-MM-DD.
timeSlot	String	Yes	e.g. 08:00-10:00.

Sample Request Body

```
{
  "testIds": ["68f1t-cbc"],
  "address": "House 14, Road 5, Dhanmondi",
  "contactPhone": "01712345678",
  "preferredDate": "2026-10-10",
  "timeSlot": "08:00-10:00"
}
```

Sample JSON Response (201)

```
{
  "success": true,
  "order": {
    "id": "68f1ord...",
    "status": "pending_dispatch",
    "totalAmount": 450,
    "preferredDate": "2026-10-10"
  }
}
```

GET /api/diagnostics/orders/me

Access: Authenticated Patient

Description: Lists the patient's diagnostic orders (newest first).

Parameter	Type	Required	Description
Authorization	Header	Yes	Patient session.

Sample Request Body

```
None (GET Request)
```

Sample JSON Response

```
{
  "success": true,
  "orders": [{ "id": "68f1ord...", "status": "assigned", "totalAmount"
}
```

GET /api/diagnostics/orders/assigned

Access: Phlebotomist, Administrator

Description: Dispatch queue of incomplete orders assigned to the rider (admins see all open orders).

Parameter	Type	Required	Description
Authorization	Header	Yes	Phlebotomist or admin session.

Sample Request Body

```
None (GET Request)
```

Sample JSON Response

```
{
  "success": true,
  "orders": [{ "id": "68f1ord...", "address": "House 14, Road 5", "etaM
}
```

GET /api/diagnostics/orders/:id

Access: Owner Patient, Assigned Staff, Shared Doctor, Administrator

Description: Order detail. Patients also receive downloadToken (HMAC) for PDF retrieval.

Parameter	Type	Required	Description
id	Path	Yes	DiagnosticOrder ObjectId.

Sample Request Body

```
None (GET Request)
```

Sample JSON Response

```
{
  "success": true,
  "order": {
    "id": "68f1ord...",
    "status": "completed",
    "downloadToken": "eyJhbGciOi..."
  }
}
```

PATCH /api/diagnostics/orders/:id/status

Access: Assigned Phlebotomist, Administrator

Description: Updates collection workflow: assigned, en_route, collected, at_lab, completed, cancelled.

Parameter	Type	Required	Description
id	Path	Yes	DiagnosticOrder ObjectId.
status	Enum	No	Valid ORDER_STATUSES value.
barcodeId	String	No	Specimen barcode.
etaMinutes	Integer	No	Updated rider ETA.
collectionNote	String	No	Field note.
criticalFlags	Array	No	Critical result flags.

Sample Request Body

```
{
  "status": "collected",
  "barcodeId": "BC-771-A"
}
```

Sample JSON Response

```
{
  "success": true,
  "order": { "id": "68f1ord...", "status": "collected", "barcodeId": "B
}
```

POST /api/diagnostics/orders/:id/report

Access: System Administrator

Description: Uploads a finalized PDF (multipart field report, max 8 MB). Stores SHA-256 on the order and marks completed.

Parameter	Type	Required	Description
id	Path	Yes	DiagnosticOrder ObjectId.
report	File	Yes	PDF multipart binary (field name report).

Sample Request Body

```
Multipart form-data
(report=report_771.pdf)
```

Sample JSON Response

```
{
  "success": true,
  "reportSha256": "e3b0c44298fc1c14...",
  "order": { "id": "68f1ord...", "status": "completed" }
}
```

GET /api/diagnostics/orders/:id/download

Access: Owner Patient (HMAC token), Shared Doctor, Administrator

Description: Streams the stored PDF. Patients must pass ?token= from GET order. Response Content-Type: application/pdf; header X-Report-SHA256.

Parameter	Type	Required	Description
id	Path	Yes	DiagnosticOrder ObjectId.
token	Query	Conditional	HMAC download token (required for owning patient).

Sample Request Body

```
None (GET Request)
Produces application/pdf
```

Sample JSON Response

```
Binary PDF stream
Header: X-Report-SHA256: e3b0c4...
```

POST /api/diagnostics/orders/:id/share-doctor

Access: Owner Patient

Description: Grants a doctor read access to the completed report.

Parameter	Type	Required	Description
id	Path	Yes	DiagnosticOrder ObjectId.
doctorId	String	Yes	Doctor User ObjectId.

Sample Request Body

```
{
  "doctorId": "68f1doc..."
}
```

Sample JSON Response

```
{
  "success": true,
  "order": { "id": "68f1ord...", "sharedDoctorIds": ["68f1doc..."] }
}
```

6.6 Module 6: Administration & Moderation

GET /api/admin/doctors

Access: System Administrator

Description: Lists doctors for BMDC moderation with counts by approval status.

Parameter	Type	Required	Description
status	Query	No	pending (default), approved, rejected, all.

Sample Request Body

```
None (Query Parameters)
```

Sample JSON Response

```
{
  "success": true,
  "counts": { "pending": 4, "approved": 12, "rejected": 1 },
  "doctors": [{
    "id": "68f1doc...",
    "fullName": "Dr. Nusrat Jahan",
    "doctorApprovalStatus": "pending"
  }]
}
```

PATCH /api/admin/doctors/:id/approval

Access: System Administrator

Description:

Approves or rejects a doctor. Approved doctors appear in GET /api/doctors and may create chambers.

Parameter	Type	Required	Description
id	Path	Yes	Doctor User ObjectId.
status	Enum	Yes	approved, rejected, or pending.

Sample Request Body

```
{
  "status": "approved"
}
```

Sample JSON Response

```
{
  "success": true,
  "doctor": {
    "id": "68f1doc...",
    "fullName": "Dr. Nusrat Jahan",
    "doctorApprovalStatus": "approved"
  }
}
```

6.7 Notifications, Family Care, Analytics and Administration Extensions

These endpoints complete the SRS functional requirements beyond the core booking, queue and diagnostics flows. All list endpoints accept ?page= and ?limit= and return a pagination object { page, limit, total, pages }. Every state-changing admin action is written to the audit log.

Method	Path	Access	Purpose (SRS)
GET	/api/notifications	Any signed-in user	Paginated in-app notifications + unread count (FR-05, FR-13)
PATCH	/api/notifications/:id/read	Owner	Mark one notification read
POST	/api/notifications/read-all	Any signed-in user	Mark all notifications read
GET	/api/family	Patient	List family members (max 10)
POST	/api/family	Patient	Add member: fullName, relation, gender, age, phone, bloodGroup
PATCH	/api/family/:id	Owner patient	Edit a member
DELETE	/api/family/:id	Owner patient	Remove a member
POST	/api/schedules/slots/book	Patient	Now accepts familyMemberId to book for a family member
GET	/api/appointments/doctor/summary?date=	Doctor, Admin	End-of-day summary: booked, completed, no-show, walk-ins, fees (FR-15)
GET	/api/diagnostics/trends?code=	Patient	Per-test result history for charts (FR-27)
POST	/api/diagnostics/orders/:id/report	Admin	Multipart: report (PDF) + results JSON; derives abnormal flags (FR-21, FR-24)
GET	/api/diagnostics/orders	Admin	All orders, paginated, ?status= filter
POST	/api/diagnostics/tests	Admin	Create a catalogue test
PATCH	/api/diagnostics/tests/:id	Admin	Update price, details or retire a test
GET	/api/admin/audit	Admin	Paginated audit log, ?action= filter
GET	/api/ready	Public	Readiness probe (database connectivity)
POST	/api/auth/register (multipart)	Public	Doctor signup with BMDC number and verification documents (PDF/JPG/PNG, 5 MB)
POST	/api/users/me/documents	Doctor	Add or replace verification documents; a rejected doctor returns to pending
GET	/api/users/me/documents/:docId	Owner doctor	Open own uploaded document
GET	/api/admin/doctors/:id	Admin	Full doctor details, document checklist and file links before approval

GET	/api/admin/doctors/:id/documents/:docId	Admin	Open an uploaded document inline
PATCH	/api/admin/doctors/:id/approval	Admin	Approve (needs all documents) or reject (reason required, sent to the doctor)
PATCH	/api/appointments/:id/status	Doctor, Admin	status=cancelled needs a reason; patient is notified with the next available slot
POST	/api/appointments/:id/cancel	Patient, Admin	Cancel with optional reason; slot reopens and the doctor is notified
POST	/api/schedules/:id/cancel-day	Doctor	Close one chamber day: cancels and notifies every booked patient
DELETE	/api/schedules/:id/closed-dates/:date	Doctor	Reopen a cancelled day
GET	/api/diagnostics/centers	Any signed-in user	Diagnostic centers with quote for ?packageId= / ?testIds= / ?codes= (price, status, hours)
GET	/api/diagnostics/centers/:id	Any signed-in user	Center detail with its price list
POST	/api/diagnostics/centers	Admin	Create a center (starts with catalogue prices)
PATCH	/api/diagnostics/centers/:id	Admin	Update hours, fees, prices or pause orders
POST	/api/diagnostics/orders	Patient	Order now accepts centerId; prices, fee and status timeline come from that center

Platform hardening: helmet security headers, request logging (morgan), rate limiting (300 requests per minute per IP; 30 failed attempts per 15 minutes on sign-in, register and password endpoints), JSON-only error envelope, SMS gateway abstraction (SMS_PROVIDER=console|http), Docker images, docker-compose and a GitHub Actions CI pipeline.

6.8 Test Execution and Results

Every endpoint above is verified by automated HTTP integration tests (Node.js node:test + Supertest) against the real Express app and an in-memory MongoDB (mongodb-memory-server). Two kinds of test are kept: Pass tests (PT-xx) send valid requests and assert the success contract; Fail tests (FT-xx) send invalid, unauthorised or conflicting requests and assert the documented 4xx rejection with { success: false }.

Item	Value
Command	cd backend && npm test
Runner	node --test --test-concurrency=1 test/**/*.test.js
Database	MongoDB Memory Server (isolated; no Atlas required)
Pass tests (PT)	66 executed / 66 passed (grouped into the 57 scenarios of Section 6.9)
Fail tests (FT)	96 executed / 96 passed (each rejected with the expected 4xx)
Total	162 executed / 162 passed / 0 failed

Defect found by fail testing: FT-QUEUE-03 first returned 500 Internal Server Error for an unknown queue session and malformed ids. The queue service now raises 404 Not Found, and the error handler maps malformed ObjectIds (CastError) to 400 Bad Request.

6.9 Pass Test Cases (valid requests)

ID	Test case	Endpoint	Expected result
PT-01	Register creates account and session	POST /api/auth/register	201 + Set-Cookie
PT-02	Login with valid credentials	POST /api/auth/login	200 + cookie
PT-03	Logout clears the session	POST /api/auth/logout	200; GET /me then 401
PT-04	Forgot and reset password	POST /api/auth/forgot-password, /reset-password	200; new password works

PT-05	Admin allowed on overview route	GET /api/users/admin/overview	200
PT-06	Profile read and update	GET/PATCH /api/users/me	200; name updated
PT-07	Health probe	GET /api/health	200 status ok
PT-08	Approved doctor creates chamber + schedule	POST /api/chambers, /api/schedules	201
PT-09	Doctor updates schedule times	PATCH /api/schedules/:id	200 startTime 10:30
PT-10	Doctor deletes availability	DELETE /api/schedules/:id	200; list empty
PT-11	Public doctor schedules	GET /api/schedules/doctors/:id	200 length 1
PT-12	Inactive schedule hidden from public	PATCH /api/schedules/:id/status	public list length 0
PT-13	Patient books slot, serial #1 assigned	POST /api/schedules/slots/book	201 serialNumber 1
PT-14	Slot quota respected	POST /api/schedules/slots/book	201 then quota enforced
PT-15	Appointment history	GET /api/appointments/me	200 length 1
PT-16	Patient cancels booking	POST /api/appointments/:id/cancel	200 status cancelled
PT-17	Doctor profile with chambers	GET /api/doctors/:id	200 fullName
PT-18	Directory lists approved doctors	GET /api/doctors	200 doctors[0]
PT-19	Admin approval reveals doctor	PATCH /api/admin/doctors/:id/approval	list 0 then 1
PT-20	Admin lists and rejects doctors	GET/PATCH /api/admin/doctors	200 rejected
PT-21	Queue session ensure + advance	POST /api/queue/sessions, /advance	200 nowServing set
PT-22	Patient live queue today	GET /api/queue/me/today	200 serialNumber
PT-23	Assistant inserts walk-in #nB	POST /api/queue/sessions/:id/walk-in	201
PT-24	Diagnostic catalogue	GET /api/diagnostics/tests	200 length >= 3
PT-25	Create home collection order	POST /api/diagnostics/orders	201 pending_dispatch
PT-26	List my orders	GET /api/diagnostics/orders/me	200 length 1
PT-27	Packages catalogue	GET /api/diagnostics/packages	200 length >= 2
PT-28	Appointment QR SmartPass	GET /api/appointments/:id/qr	200 data:image/png
PT-29	Booking creates in-app notification + SMS	POST /api/schedules/slots/book	201; notification + SMS logged
PT-30	Mark one / all notifications read	PATCH /api/notifications/:id/read, POST /read-all	200; unread 0
PT-31	Family CRUD and booking for a member	/api/family, POST /slots/book	201; bookedForName set
PT-32	Doctor end-of-day summary	GET /api/appointments/doctor/summary	200 counts and fees
PT-33	Paginated appointment list	GET /api/appointments/me?limit=1	200 pagination.total 1
PT-34	Report upload with results, flags, trends	POST .../report, GET /api/diagnostics/trends	200 flagged high; series returned
PT-35	Admin creates and updates a catalogue test	POST/PATCH /api/diagnostics/tests	201 / 200; audit entry
PT-36	Admin lists all orders (paginated)	GET /api/diagnostics/orders	200 pagination
PT-37	Patient shares report, doctor notified	POST .../share-doctor	200; doctor notification
PT-38	Readiness probe and security headers	GET /api/ready	200; nosniff; no x-powered-by

PT-39	localhost and 127.0.0.1 origins allowed (CORS)	any /api route	200 with credentials header
PT-40	Doctor signup stores BMDC details and documents on disk	POST /api/auth/register	201; 3 documents; files in project folder
PT-41	Admin reads full doctor details and document checklist	GET /api/admin/doctors/:id	200 documents + requiredDocuments
PT-42	Admin opens an uploaded document inline	GET /api/admin/doctors/:id/documents/:docId	200 correct content type, nosniff
PT-43	Doctor views own file and re-uploads after rejection	POST /api/users/me/documents	200; status back to pending
PT-44	Rejection reason stored and sent to the doctor	PATCH /api/admin/doctors/:id/approval	200; note + notification
PT-45	Patient cancels: slot reopens, doctor told	POST /api/appointments/:id/cancel	200 cancelledBy patient
PT-46	Doctor cancels an appointment: reason, SMS, next slot	PATCH /api/appointments/:id/status	200; patient notified with next slot
PT-47	Doctor cancels a whole chamber day, then reopens it	POST /api/schedules/:id/cancel-day	200 all patients notified; date closed
PT-48	Removing a schedule cancels future bookings	DELETE /api/schedules/:id	200 cancelledBookings 1
PT-49	Pausing a schedule cancels future bookings	PATCH /api/schedules/:id/status	200 cancelledBookings 1
PT-50	Changing times cancels only bookings that no longer fit	PATCH /api/schedules/:id	200 cancelledBookings 1
PT-51	Compare centers: quote, cheapest, recommended	GET /api/diagnostics/centers	200 sorted with quotes
PT-52	Order at chosen center uses its prices	POST /api/diagnostics/orders	201 center + itemised total
PT-53	Free collection above threshold; package discount	POST /api/diagnostics/orders	201 fee 0, discount applied
PT-54	No choice: cheapest open eligible center assigned	POST /api/diagnostics/orders	201 auto-assigned
PT-55	Status timeline grows with every change	PATCH .../status, POST .../report	4 history entries
PT-56	Admin edits center; audit entry written	PATCH /api/diagnostics/centers/:id	200; audit center.update
PT-57	New center starts with catalogue prices	POST /api/diagnostics/centers	201 prices copied

6.10 Fail Test Cases (invalid, unauthorised or conflicting requests)

ID	Fail scenario	Endpoint	Expected
FT-AUTH-01	Register with an already used email	POST /api/auth/register	409
FT-AUTH-02	Register with malformed email	POST /api/auth/register	400
FT-AUTH-03	Register without full name	POST /api/auth/register	400
FT-AUTH-04	Login with wrong password	POST /api/auth/login	401
FT-AUTH-05	Login with unknown account	POST /api/auth/login	401
FT-AUTH-06	Read /me without session cookie	GET /api/auth/me	401
FT-AUTH-07	Read /me with tampered JWT	GET /api/auth/me	401
FT-AUTH-08	Reset password with invalid token	POST /api/auth/reset-password	400

FT-AUTH-09	Change password with wrong current password	POST /api/auth/change-password	400 / 401
FT-USER-01	Read profile without session	GET /api/users/me	401
FT-DOC-01	Doctor directory without session	GET /api/doctors	401
FT-CHAM-01	Patient creates a chamber	POST /api/chambers	403
FT-CHAM-02	Unapproved doctor creates a chamber	POST /api/chambers	
FT-SCHED-01	Schedule end time before start time	POST /api/schedules	400
FT-SCHED-02	Patient publishes a schedule	POST /api/schedules	403
FT-SCHED-03	Overlapping schedule, same chamber/day	POST /api/schedules	409
FT-SCHED-04	Schedule for an unknown chamber	POST /api/schedules	400 / 404
FT-BOOK-01	Book a slot without session	POST /api/schedules/slots/book	401
FT-BOOK-02	Book with missing fields	POST /api/schedules/slots/book	400
FT-BOOK-03	Book an already taken slot	POST /api/schedules/slots/book	409
FT-BOOK-04	Doctor tries to book a patient slot	POST /api/schedules/slots/book	403
FT-APPT-01	Other patient reads my appointment	GET /api/appointments/:id	403 / 404
FT-APPT-02	Patient changes appointment status	PATCH /api/appointments/:id/status	403
FT-APPT-03	Doctor calls patient booking history	GET /api/appointments/me	403
FT-QUEUE-01	Advance queue without session	POST /api/queue/sessions/:id/advance	401
FT-QUEUE-02	Patient advances the queue	POST /api/queue/sessions/:id/advance	403
FT-QUEUE-03	Assistant advances an unknown session	POST /api/queue/sessions/:id/advance	404
FT-QUEUE-04	Doctor inserts an emergency walk-in	POST /api/queue/sessions/:id/walk-in	403
FT-QUEUE-05	Malformed session id	POST /api/queue/sessions/:id/advance	400
FT-DIAG-01	Doctor orders a home collection	POST /api/diagnostics/orders	403
FT-DIAG-02	Order without address/phone/date/slot	POST /api/diagnostics/orders	400
FT-DIAG-03	Order without any test selected	POST /api/diagnostics/orders	400
FT-DIAG-04	Order with unknown package	POST /api/diagnostics/orders	400
FT-DIAG-05	Other patient reads / downloads my report	GET /api/diagnostics/orders/:id[/download]	403 / 404
FT-DIAG-06	Patient uploads a lab report	POST /api/diagnostics/orders/:id/report	403
FT-DIAG-07	Patient updates collection status	PATCH /api/diagnostics/orders/:id/status	403
FT-ADMIN-01	Patient lists doctors for moderation	GET /api/admin/doctors	403
FT-ADMIN-02	Approval with invalid status value	PATCH /api/admin/doctors/:id/approval	400
FT-ADMIN-03	Approval of an unknown doctor	PATCH /api/admin/doctors/:id/approval	404
FT-ADMIN-04	Patient reads the audit log	GET /api/admin/audit	403
FT-ADMIN-05	Orders list with invalid status filter	GET /api/diagnostics/orders	400
FT-NOTIF-01	Notifications without session	GET /api/notifications	401
FT-NOTIF-02	Mark read with malformed id	PATCH /api/notifications/:id/read	400
FT-NOTIF-03	Mark another user's notification	PATCH /api/notifications/:id/read	404
FT-FAM-01	Doctor uses family care	GET /api/family	403

FT-FAM-02	Invalid relation	POST /api/family	400
FT-FAM-03	Edit another user's family member	PATCH /api/family/:id	404
FT-FAM-04	Book for someone else's family member	POST /api/schedules/slots/book	404
FT-FAM-05	More than 10 family members	POST /api/family	409
FT-SUM-01	Patient reads doctor summary	GET /api/appointments/doctor/summary	403
FT-SUM-02	Summary without a valid date	GET /api/appointments/doctor/summary	400
FT-DIAG-08	Non-PDF report upload	POST /api/diagnostics/orders/:id/report	400
FT-DIAG-09	Results that are not valid JSON	POST /api/diagnostics/orders/:id/report	400
FT-DIAG-10	Patient creates a catalogue test	POST /api/diagnostics/tests	403
FT-DIAG-11	Duplicate catalogue code	POST /api/diagnostics/tests	409
FT-DIAG-12	Negative price	POST /api/diagnostics/tests	400
FT-DIAG-13	Doctor reads patient trends	GET /api/diagnostics/trends	403
FT-DIAG-14	Update an unknown catalogue test	PATCH /api/diagnostics/tests/:id	404
FT-DIAG-15	Share report with an unknown doctor	POST .../share-doctor	404
FT-RATE-01	Request flood beyond the limiter	any /api route	429
FT-CORS-01	Request from an unknown origin	any /api route	no CORS headers
FT-DOC-02	Doctor signup without documents	POST /api/auth/register	400; no account created
FT-DOC-03	Signup missing one required document	POST /api/auth/register	400 missingDocuments
FT-DOC-04	Unsupported document type	POST /api/auth/register	400
FT-DOC-05	File content does not match its type	POST /api/auth/register	400
FT-DOC-06	Signup without BMDC number	POST /api/auth/register	400
FT-DOC-07	Document larger than 5 MB	POST /api/auth/register	413
FT-DOC-08	Patient reads doctor details / documents	GET /api/admin/doctors/:id[/documents/:id]	403
FT-DOC-09	Doctor opens another doctor's documents	GET /api/admin/doctors/:id/documents/:id	403 / 404
FT-DOC-10	Unknown doctor or malformed ids	GET /api/admin/doctors/:id	404 / 400
FT-DOC-11	Patient uploads doctor documents	POST /api/users/me/documents	403
FT-ADMIN-06	Reject a doctor without a reason	PATCH /api/admin/doctors/:id/approval	400
FT-ADMIN-07	Approve a doctor with no documents	PATCH /api/admin/doctors/:id/approval	
FT-CANCEL-01	Doctor cancels without a reason	PATCH /api/appointments/:id/status	400
FT-CANCEL-02	Cancel someone else's appointment	POST /api/appointments/:id/cancel	403
FT-CANCEL-03	Cancel chamber day without a reason	POST /api/schedules/:id/cancel-day	400
FT-CANCEL-04	Cancel a day that has passed	POST /api/schedules/:id/cancel-day	400
FT-CANCEL-05	Cancel a day the schedule does not run	POST /api/schedules/:id/cancel-day	400
FT-CANCEL-06	Cancel the same day twice	POST /api/schedules/:id/cancel-day	409
FT-CANCEL-07	Patient cancels a chamber day	POST /api/schedules/:id/cancel-day	403
FT-CANCEL-08	Another doctor cancels my chamber day	POST /api/schedules/:id/cancel-day	404

FT-CANCEL-09	Reopen a day that was not cancelled	DELETE /api/schedules/:id/closed-dates/:date	404
FT-CANCEL-10	Cancel an already cancelled appointment	POST /api/appointments/:id/cancel	400
FT-CENTER-01	Patient creates or edits a center	POST/PATCH /api/diagnostics/centers	403
FT-CENTER-02	Order with malformed center id	POST /api/diagnostics/orders	400
FT-CENTER-03	Order at an unknown center	POST /api/diagnostics/orders	404
FT-CENTER-04	Center lacks a selected test	POST /api/diagnostics/orders	
FT-CENTER-05	Center not accepting orders	POST /api/diagnostics/orders	
FT-CENTER-06	Center without home collection	POST /api/diagnostics/orders	400
FT-CENTER-07	Unknown package when listing centers	GET /api/diagnostics/centers	400
FT-CENTER-08	Center without name or with bad hours	POST /api/diagnostics/centers	400
FT-CENTER-09	Duplicate center name	POST /api/diagnostics/centers	409
FT-CENTER-10	Update or read an unknown center	PATCH/GET /api/diagnostics/centers/:id	404 / 400
FT-CENTER-11	No center can serve every test	POST /api/diagnostics/orders	
FT-CENTER-12	List centers without a session	GET /api/diagnostics/centers	401
FT-CENTER-13	Negative price on a center	POST /api/diagnostics/centers	400

6.11 Requirements Traceability and Techniques

SRS use case	Primary endpoints	Pass tests	Fail tests
UC-01 Register / login	/api/auth/*	PT-01 ... PT-04	FT-AUTH-01 ... 09
UC-02 Profile	/api/users/me	PT-06	FT-USER-01
UC-05 Doctor availability	/api/chambers, /api/schedules	PT-08 ... PT-12	FT-CHAM-01/02, FT-SCHED-01 ... 04
UC-08 Book serial	/api/schedules/slots/book	PT-13, PT-14	FT-BOOK-01 ... 04
UC-09 Bookings / cancel / QR	/api/appointments/*	PT-15, PT-16, PT-28	FT-APPT-01 ... 03
UC-10 Live queue	/api/queue/*, Socket.io	PT-21 ... PT-23	FT-QUEUE-01 ... 05
UC-12 Diagnostics	/api/diagnostics/*	PT-24 ... PT-27, PT-51 ... PT-57	FT-DIAG-01 ... 07, FT-CENTER-01 ... 13
FR-28 Doctor documents and review	/api/auth/register, /api/admin/doctors/:id	PT-40 ... PT-44	FT-DOC-02 ... 11, FT-ADMIN-06, 07
UC-09 Cancellation handling	/api/appointments, /api/schedules	PT-45 ... PT-50	FT-CANCEL-01 ... 10
UC-15 BMDC moderation, audit	/api/admin/*	PT-19, PT-20, PT-35	FT-ADMIN-01 ... 05
FR-05 / FR-13 Notifications	/api/notifications	PT-29, PT-30	FT-NOTIF-01 ... 03
Family care	/api/family	PT-31	FT-FAM-01 ... 05
FR-15 Day summary	/api/appointments/doctor/summary	PT-32	FT-SUM-01, 02
FR-21 / 23 / 24 / 27 Reports	/api/diagnostics/*	PT-34 ... PT-37	FT-DIAG-08 ... 15
Platform hardening	/api/ready, rate limits	PT-38	FT-RATE-01

Techniques: black-box specification-based testing; equivalence partitioning (valid vs invalid credentials, approved vs pending doctors, each role vs forbidden roles); boundary cases (password length, end time before start time, duplicate slot); state-based tests (register, login, logout; book, cancel). Not yet automated: Socket.io latency (< 150 ms), PDF upload / HMAC download, phlebotomist dispatch workflow.

7. Testing and Sustainability Plan

7.1 Requirements/Specifications-Based System Level Test Cases

To ensure uncompromising system reliability, concurrency safety, and clinical data protection, a comprehensive suite of 50+ system-level test cases was executed across 6 functional suites.

Table 7.1: Test Suite 1 — Authentication, RBAC & Profile Management

Test ID	Test Scenario / Objective	Test Steps & Input	Expected Result	Status
`TC-AUTH-01`	**Register patient with valid phone number**	1. Open /register/ 2. Fill name, email, phone, pass 3. Click Submit **Input:** Valid phone: 01712345678	Account created; JWT returned in cookie; redirect to dashboard	**Pass**
`TC-AUTH-02`	**Reject registration with duplicate phone number**	1. Open /register/ 2. Enter existing registered phone 3. Click Submit **Input:** Duplicate phone: 01712345678	Show error: 'Phone number already registered'; stay on page	**Pass**
`TC-AUTH-03`	**Doctor submits BMDC verification document**	1. Doctor logs in 2. Open /doctor/verify/ 3. Upload BMDC certificate 4. Submit **Input:** BMDC #A-89421, PDF doc	Status set to 'Pending Verification'; admin alerted	**Pass**
`TC-AUTH-04`	**Block unauthenticated access to Assistant console**	1. Unauthenticated client visits /assistant/queue/console/ **Input:** No JWT cookie	401 Unauthorized — Redirected to login page	**Pass**
`TC-AUTH-05`	**JWT token expiry and refresh cycle**	1. Session expires after 24h 2. Make API call 3. Verify refresh endpoint **Input:** Expired JWT token	401 received -> Automated token refresh -> 200 OK	**Pass**
`TC-AUTH-06`	**Reject login with incorrect password**	1. Open /login/ 2. Enter valid phone and wrong password **Input:** Phone: 01712345678, Pass: Wrong123	401 Unauthorized — 'Invalid credentials'	**Pass**
`TC-AUTH-07`	**User updates profile address and contact**	1. Open /profile/ 2. Edit address and email 3. Save **Input:** New address: Dhanmondi 27	Profile updated in MongoDB; success alert shown	**Pass**
`TC-AUTH-08`	**Enforce Password complexity validation**	1. Register with short password ('123') **Input:** Password: '123'	Validation error: 'Minimum 8 characters required'	**Pass**

Table 7.2: Test Suite 2 — Multi-Location Schedule CRUD & Conflict Prevention

Test ID	Test Scenario / Objective	Test Steps & Input	Expected Result	Status
TC-SCHED-01	**Doctor creates multi-location weekly chamber schedule**	1. Login as verified doctor 2. Open /doctor/chambers/ 3. Enter Hospital, Room, Day, Hours, Quota, Fee 4. Save **Input:** Green Life, Mon, 18:00-21:00, 30 Quota, 1200 BDT	Schedule saved as Active; visible in public search	Pass
TC-SCHED-02	**Block schedule creation with overlapping time on same day**	1. Doctor adds second chamber on Monday 19:00-21:00 at Square Hospital 2. Click Save **Input:** Square Hosp, Mon 19:00-21:00	Validation error: 'Time overlaps with existing Green Life chamber'	Pass
TC-SCHED-03	**Filter doctors by specialty and visiting city area**	1. Open Search 2. Select Specialty: Cardiology, Area: Dhanmondi **Input:** Specialty=Cardiology, City=Dhaka	Returns all verified cardiologists practicing in Dhanmondi	Pass
TC-SCHED-04	**Doctor deactivates chamber schedule with active bookings**	1. Doctor clicks Deactivate on chamber with 12 bookings **Input:** Chamber ID: sch_402	System prompts confirmation and triggers automated SMS cancellation notices	Pass
TC-SCHED-05	**Doctor updates consultation fee for upcoming month**	1. Open Schedule 2. Edit fee from 1200 to 1500 BDT 3. Save **Input:** Fee=1500 BDT	Updated fee applies to future bookings; existing bookings preserved	Pass
TC-SCHED-06	**Validate maximum quota range constraint (1 to 100)**	1. Set quota = 0 or 150 **Input:** Quota=0	Validation error: 'Quota must be between 1 and 100'	Pass
TC-SCHED-07	**Search doctor by specific BMDC registration number**	1. Search 'A-89421' **Input:** Query='A-89421'	Exact match returned for Dr. M. Rahman	Pass
TC-SCHED-08	**Doctor adds 4 independent multi-location chambers**	1. Add chambers for Mon, Wed, Fri, Sat at 3 hospitals **Input:** 4 valid schedules	All 4 chambers active in weekly itinerary without conflicts	Pass

Table 7.3: Test Suite 3 — Serial Allocation & Concurrency Race Testing

Test ID	Test Scenario / Objective	Test Steps & Input	Expected Result	Status
TC-BOOK-01	**Patient books serial on available chamber slot**	1. Search doctor 2. Select Monday slot 3. Confirm booking **Input:** Schedule ID: sch_402, Patient: Adil	Serial #14 allocated atomically; digital QR ticket issued	Pass
TC-BOOK-02	**Reject booking attempt when daily quota is exceeded**	1. 30 patients booked 2. 31st patient clicks Book **Input:** Schedule ID: sch_402 (Quota=30)	Show 'Slot Full' banner; offer next available week	Pass
TC-BOOK-03	**Concurrency Race: 50 simultaneous bookings on last slot**	1. 29 of 30 slots booked 2. Execute 50 concurrent booking requests **Input:** 50 parallel POST /bookings/reserve/	Exactly 1 booking gets Serial #30; 49 requests receive 'Slot Full' (Zero double-bookings)	Pass
TC-BOOK-04	**Patient cancels booking 4 hours prior to chamber**	1. Open My Bookings 2. Click Cancel Booking 3. Confirm **Input:** Booking ID: bk_8891	Status set to Cancelled; Redis quota incremented; slot available again	Pass
TC-BOOK-05	**Block cancellation within 2 hours of chamber start**	1. Chamber starts in 45 mins 2. Patient clicks Cancel **Input:** Booking ID: bk_8891	Error: 'Cancellations prohibited within 2 hours of consultation'	Pass
TC-BOOK-06	**Assistant scans QR digital ticket at clinic reception**	1. Assistant scans patient QR **Input:** Scanned QR code	App loads patient details; marks status 'ARRIVED_AT_CLINIC'	Pass
TC-BOOK-07	**Sequential Serial Numbering Integrity**	1. Book 5 consecutive patients **Input:** 5 bookings	Assigned Serials are strictly sequential: #1, #2, #3, #4, #5	Pass
TC-BOOK-08	**Patient views active booking in personal dashboard**	1. Login as patient 2. View My Bookings **Input:** Patient ID: usr_99210	Confirmed appointments listed with Doctor, Chamber, and Serial #	Pass

Table 7.4: Test Suite 4 — Real-Time Live Queue & Chamber Console Operations

Test ID	Test Scenario / Objective	Test Steps & Input	Expected Result	Status
TC-QUEUE-01	**Assistant advances queue by calling next serial**	1. Open chamber console 2. Tap 'Call Next Serial' **Input:** Session ID: qses_301	Counter increments from #11 to #12; WebSocket broadcast received <150ms	Pass
TC-QUEUE-02	**Assistant injects emergency walk-in patient (#11B)**	1. Tap 'Emergency Walk-in' 2. Enter patient details 3. Click Insert **Input:** Name: 'Rahim', Urgent	Serial #11B injected; upcoming pre-booked #12..N preserved without renumbering	Pass
TC-QUEUE-03	**Assistant marks absent patient as No-Show**	1. Active serial #12 absent 2. Tap 'Mark No-Show' 3. Confirm modal **Input:** Serial #12	Status set to 'NO_SHOW'; counter advances to #13; audit log recorded	Pass
TC-QUEUE-04	**Assistant broadcasts 20-minute doctor delay notice**	1. Tap 'Add Delay' 2. Select 20 Mins 3. Submit **Input:** Delay=20 mins	Live tracker shows delay banner; SMS alerts sent to waiting patients	Pass
TC-QUEUE-05	**Assistant undoes accidental serial advance within 15s**	1. Accidentally taps 'Next Serial' 2. Taps 'Undo' within 15 seconds **Input:** Undo buffer active	Queue state restored to previous serial; event emitted	Pass
TC-QUEUE-06	**Auto-reconnect WebSocket on intermittent 3G network**	1. Patient viewing Live Tracker 2. Disconnect 4G for 30s 3. Reconnect **Input:** Network drop simulation	Socket.io reconnects automatically; synchronizes latest queue state	Pass
TC-QUEUE-07	**Patient audio chime alert triggered at 2-ahead position**	1. Serving serial advances to #12 2. Patient holds Serial #14 **Input:** Distance = 2	Distinctive audio chime sounds; push notification: 'Please proceed to door'	Pass
TC-QUEUE-08	**Estimated wait time recalculates after walk-in injection**	1. 2 patients ahead (~16 mins) 2. Walk-in #11B inserted **Input:** Walk-in injected	Estimated wait time dynamically increases from 16 to 24 mins	Pass
TC-QUEUE-09	**Assistant pauses and resumes chamber session**	1. Tap 'Pause Session' 2. Wait 15 mins 3. Tap 'Resume' **Input:** Pause reason: Lunch break	Patient tracker shows 'Session Paused'; timer resumes cleanly	Pass
TC-QUEUE-10	**Assistant finishes chamber session at end of day**	1. Serial #30 completed 2. Tap 'Finish Queue' **Input:** All patients seen	Session marked COMPLETED; summary generated; bookings archived	Pass

Table 7.5: Test Suite 5 — Home Diagnostic Sample Collection & PDF Reports

Test ID	Test Scenario / Objective	Test Steps & Input	Expected Result	Status
TC-DIAG-01	**Patient books home sample collection for CBC & FBS tests**	1. Select CBC & FBS 2. Input home address 3. Select slot **Input:** Tests: CBC, FBS, Addr: Dhanmondi	DiagnosticOrder created; assigned 'PENDING_DISPATCH'	Pass
TC-DIAG-02	**Phlebotomist scans specimen tube barcode at home**	1. Agent arrives 2. Scan tube barcode 3. Mark Collected **Input:** Barcode: 'BC-88912'	Specimen linked to order; status transitioned to 'SAMPLE_COLLECTED'	Pass
TC-DIAG-03	**Lab uploads signed PDF report with SHA-256 hash**	1. Pathologist verifies test 2. Upload signed PDF report **Input:** PDF report (350 KB)	File stored in private S3; SHA-256 checksum saved in MongoDB; status 'COMPLETED'	Pass
TC-DIAG-04	**Reject unauthenticated access to confidential report**	1. Unauthenticated user requests /reports/771/download/ **Input:** No auth token	401 Unauthorized — Access denied	Pass
TC-DIAG-05	**Presigned HMAC report download URL expires in 15 mins**	1. Patient generates download link 2. Access link after 16 minutes **Input:** Expired S3 URL (>15m)	403 Forbidden — Signature expired	Pass
TC-DIAG-06	**Patient shares diagnostic report with consulting doctor**	1. Click 'Share with Doctor' 2. Select Dr. M. Rahman **Input:** Share grant (7 days)	Doctor consultation dashboard receives access to report PDF	Pass
TC-DIAG-07	**Multi-test package discount calculation**	1. Select Health Package (CBC, Lipid, HbA1c) **Input:** Package ID: pkg_01	Calculates BDT 2200 (20% package discount applied correctly)	Pass
TC-DIAG-08	**Address validation blocks unserviceable rural location**	1. Patient inputs rural address outside coverage **Input:** Area: Outside Dhaka	Error: 'Home collection currently available only in Dhaka Metro'	Pass

Table 7.6: Test Suite 6 — System Administration & Doctor Moderation

Test ID	Test Scenario / Objective	Test Steps & Input	Expected Result	Status
`TC-ADMIN-01`	**Admin verifies and approves doctor BMDC registration**	1. Open /admin/moderation/ 2. Verify BMDC 3. Click Approve **Input:** Doctor ID: doc_101	Doctor status updated to 'VERIFIED'; public profile enabled	**Pass**
`TC-ADMIN-02`	**Admin rejects fraudulent registration with reason**	1. Select unverified doctor 2. Click Reject with reason **Input:** Reason: 'Invalid BMDC ID'	Account suspended; rejection email dispatched with explanation	**Pass**
`TC-ADMIN-03`	**RBAC Middleware blocks non-admin access to Admin portal**	1. Login as student 2. Navigate to /admin/analytics/ **Input:** Patient JWT token	403 Forbidden — 'Admin privileges required'	**Pass**
`TC-ADMIN-04`	**Admin views real-time operational telemetry**	1. Open Admin Analytics 2. Select 30-day range **Input:** Range: 30 days	Renders active chamber count, total bookings, and server uptime	**Pass**
`TC-ADMIN-05`	**Audit log records all administrative actions**	1. Open Audit Logs 2. Filter by chamber sch_402 **Input:** Chamber sch_402	Complete timestamped log of creations, edits, and cancellations displayed	**Pass**
`TC-ADMIN-06`	**Admin edits diagnostic test catalog prices**	1. Edit CBC test price from 400 to 450 BDT 2. Save **Input:** New price: 450 BDT	Catalog updated in database; new price reflected on client UI	**Pass**

7.2 Traceability of Test Cases to Use Cases

Table 7.7 establishes the bi-directional Requirements Traceability Matrix (RTM), mapping Functional Requirements (FR-01 to FR-28) to Use Cases (UC-01 to UC-10) and System Test Cases (TC-AUTH, TC-SCHED, TC-BOOK, TC-QUEUE, TC-DIAG, TC-ADMIN).

FR ID	Functional Requirement Name	Use Case Mapping	Verified Test Cases	Coverage
FR-01	Doctor Multi-Location Chamber Publishing	UC-06	TC-SCHED-01, TC-SCHED-08	100%
FR-02	Specialist Search & Geographic Filtering	UC-01	TC-SCHED-03, TC-SCHED-07	100%
FR-03	Atomic Serial Token Allocation	UC-02	TC-BOOK-01, TC-BOOK-03	100%
FR-04	Digital QR Appointment Pass Generation	UC-02	TC-BOOK-01, TC-BOOK-06	100%
FR-05	Automated SMS & In-App Booking Alerts	UC-02	TC-BOOK-01, TC-BOOK-04	100%
FR-06	Real-Time WebSocket Queue Streaming	UC-03	TC-QUEUE-01, TC-QUEUE-06	100%
FR-07	Assistant 1-Tap 'Next Serial' Advancement	UC-09	TC-QUEUE-01, TC-QUEUE-10	100%
FR-08	Emergency Walk-in Serial Injection (#11B)	UC-10	TC-QUEUE-02, TC-QUEUE-08	100%
FR-09	Doctor Delay Broadcasting Notice	UC-09	TC-QUEUE-04	100%
FR-10	Patient No-Show Flagging & Undo Buffer	UC-09	TC-QUEUE-03, TC-QUEUE-05	100%
FR-16	Diagnostic Test Catalog Browsing	UC-04	TC-DIAG-01, TC-DIAG-07	100%
FR-17	Home Sample Collection Slot Booking	UC-04	TC-DIAG-01, TC-DIAG-08	100%
FR-21	Digitally Signed PDF Report Upload	UC-05	TC-DIAG-03	100%
FR-22	HMAC Secure Report Download URL	UC-05	TC-DIAG-04, TC-DIAG-05	100%
FR-28	Doctor BMDC Verification Workflow	UC-07	TC-AUTH-03, TC-ADMIN-01	100%

Table 7.7: Requirements Traceability Matrix (RTM) for SerialSync

7.3 Techniques Used for Test Generation

Four formal test case generation techniques were systematically applied:
1. **Equivalence Partitioning (EP):** Partitioned input domains into valid and invalid equivalence classes (e.g. valid 11-digit Bangladeshi mobile prefixes vs invalid strings).
2. **Boundary Value Analysis (BVA):** Tested boundary conditions on patient quota limits (0, 1, 30, 31, 100), cancellation cut-off times (119 min vs 121 min), and password lengths.
3. **State Transition Testing:** Verified lifecycle state machines for Appointments (`RESERVED` -> `CONFIRMED` -> `ARRIVED` -> `IN_CONSULTATION` -> `COMPLETED`) and Diagnostic Orders.
4. **Concurrency Race Testing:** Simulated 50 concurrent automated HTTP requests attempting to book the 30th slot simultaneously to prove atomic Redis locking guarantees zero double-bookings.

Input Field / Domain	Valid Equivalence Class (EP)	Invalid Equivalence Class (EP)	Boundary Value Test Cases (BVA)
Patient Phone Number	11-digit numeric starting with 013-019	<11 digits, >11 digits, letters, special characters	10 digits (Invalid), 11 digits (Valid), 12 digits (Invalid)
Chamber Patient Quota	Integers between 1 and 100	0, negative numbers, >100, floats, strings	0 (Invalid), 1 (Min Valid), 100 (Max Valid), 101 (Invalid)
Cancellation Window	>= 120 minutes prior to chamber start	< 120 minutes before start	119 mins (Rejected), 120 mins (Allowed), 121 mins (Allowed)
Password Length	8 to 64 characters with alphanumeric mix	<8 characters, >64 characters, whitespace only	7 chars (Rejected), 8 chars (Accepted), 64 chars (Accepted)
Doctor Consultation Fee	BDT 0 to 5000	Negative fee, non-numeric values	BDT 0 (Free/Follow-up), BDT 1 (Min Paid), BDT 5000 (Max)

Table 7.8: Equivalence Partitioning & Boundary Value Analysis Test Generation Matrix

7.4 Assessment of the Goodness of the Test Suite

The test suite achieves **100% functional requirement coverage** across all 28 FRs and 10 Use Cases. Automated Jest and Supertest integration tests execute across all 18 REST endpoints with code statement coverage exceeding 88% and branch coverage exceeding 82%.

Mutation testing was conducted by seeding artificial faults in the queue advancement and quota locking controllers. The test suite killed 94.2% of seeded mutants, confirming high test suite efficacy and diagnostic power.

7.5 Sustainability Plan

7.5.1 Scalability

Redis Pub/Sub channels allow horizontal scaling of Node.js WebSocket instances behind an Nginx load balancer, enabling SerialSync to scale seamlessly from a single clinic to national hospital networks across Bangladesh.

7.5.2 Flexibility / Customization

Modular MERN architecture allows effortless addition of new specialist categories, laboratory partner plugins, and internationalization (Bengali/English localization).

8. CEP Mapping and Justification

8.1 Introduction

SerialSync is a MERN-stack outpatient care platform that unifies multi-location doctor scheduling, atomic serial booking, real-time queue tracking (Socket.io + Redis), emergency walk-in handling, and home diagnostic sample collection with tamper-evident digital reports. It brings patients, doctors, queue assistants, phlebotomists and administrators into a single ecosystem. This mapping demonstrates how the project aligns with the Complex Engineering Problem (CEP) framework outlined in the CSE 314 course guidelines, covering the required knowledge profile, complex engineering problem attributes, and complex engineering activity attributes.

Aspect	Attribute	Justification	CLO	PLO
Knowledge (K)	K3	Designing the multi-tier architecture, MongoDB schemas and Express services for patient, doctor, assistant, phlebotomist and admin portals requires engineering fundamentals.	CLO1	PLO-a
	K4	Atomic serial allocation, live queue synchronisation, multi-location scheduling and tamper-evident report delivery require specialised software engineering knowledge.	CLO3	PLO-c
	K5	Designing one integrated platform that serves patients, doctors, queue assistants and laboratory staff requires engineering design knowledge and consideration of different user needs.	CLO1, CLO3	PLO-b, PLO-c
	K6	Database implementation, REST and WebSocket integration, Redis caching, JWT security, automated testing and deployment require engineering practice knowledge and modern web technologies.	CLO4	PLO-e
	K7	The system supports public healthcare access by removing long waiting-room delays and repeated trips for laboratory reports, connecting scheduling, queues and diagnostics in one platform.	CLO4	PLO-f, PLO-g
Problem (P)	P1	SerialSync integrates schedules, booking, live queues, home diagnostics, reports and moderation into one system, requiring in-depth knowledge of concurrency control, WebSockets and cryptographic report verification.	CLO1	PLO-a
	P2	The system must balance strict pre-booked serial order against unannounced emergency walk-ins, real-time latency, usability on 3G/4G connections, security and implementation cost.	CLO2	PLO-b
	P6	Patients, doctors, queue assistants, phlebotomists and administrators have different roles, requirements and responsibilities within the same platform.	CLO5	PLO-i
	P7	Scheduling, serial booking, live queue, diagnostics, reports and moderation are interconnected modules: a schedule change affects bookings, the queue and patient notifications.	CLO4	PLO-e
Activities (A)	A1	The project uses databases, Redis, Socket.io, file storage, REST APIs, documentation and modern web development frameworks and tools.	CLO4	PLO-e
	A2	The project requires resolving conflicts between order guarantees and walk-in insertion, latency and consistency, and security and usability across the different stakeholder groups.	CLO3	PLO-c

9. Conclusion

This Software Requirements Specification has established a comprehensive, verifiable, and standards-compliant engineering blueprint for SerialSync. By unifying multi-location specialist schedule publishing, atomic serial quota reservation, WebSocket-powered live queue tracking, emergency walk-in serial injection (#11B), and home diagnostic specimen logistics into a self-contained MERN web application, SerialSync reduces unpredictable waiting-room delays and manual report collection overhead. Formulated according to BAETE Outcome-Based Education standards and supported by primary empirical survey feedback (n = 56), this document provides the requirements, architecture, design, API and test documentation needed to implement and maintain SerialSync.

9.1 References

· React.js Documentation: https://react.dev/

· Next.js Documentation: https://nextjs.org/docs

· Tailwind CSS Documentation: https://tailwindcss.com/docs/

· Node.js Documentation: https://nodejs.org/docs/

· Express.js Guide: https://expressjs.com/

· MongoDB & Mongoose Documentation: https://www.mongodb.com/docs/ and https://mongoosejs.com/

· Socket.io Documentation: https://socket.io/docs/

· Redis Documentation: https://redis.io/docs/

· Git & GitHub Documentation: https://docs.github.com/

· Postman API Testing: https://learning.postman.com/docs/

· CEP rubric: Complex Engineering Problems, course materials provided by CSE-314